

20. I/O Ports

20.1 Overview

The I/O port pins operate as general I/O port pins, I/O pins for peripheral modules, interrupt input pins, analog I/O, port group function for the ELC or bus control pins.

All pins except P209 (as TDO of JTAG ports) operate as input pins immediately after a reset, and pin functions are switched by register settings. The I/O ports and peripheral modules for each pin are specified in the associated registers.

Figure 20.1 shows a connection diagram for the I/O port registers. The configuration of the I/O ports differs for different packages. Table 20.1 lists the I/O port specifications by package, and Table 20.2 lists the port functions.

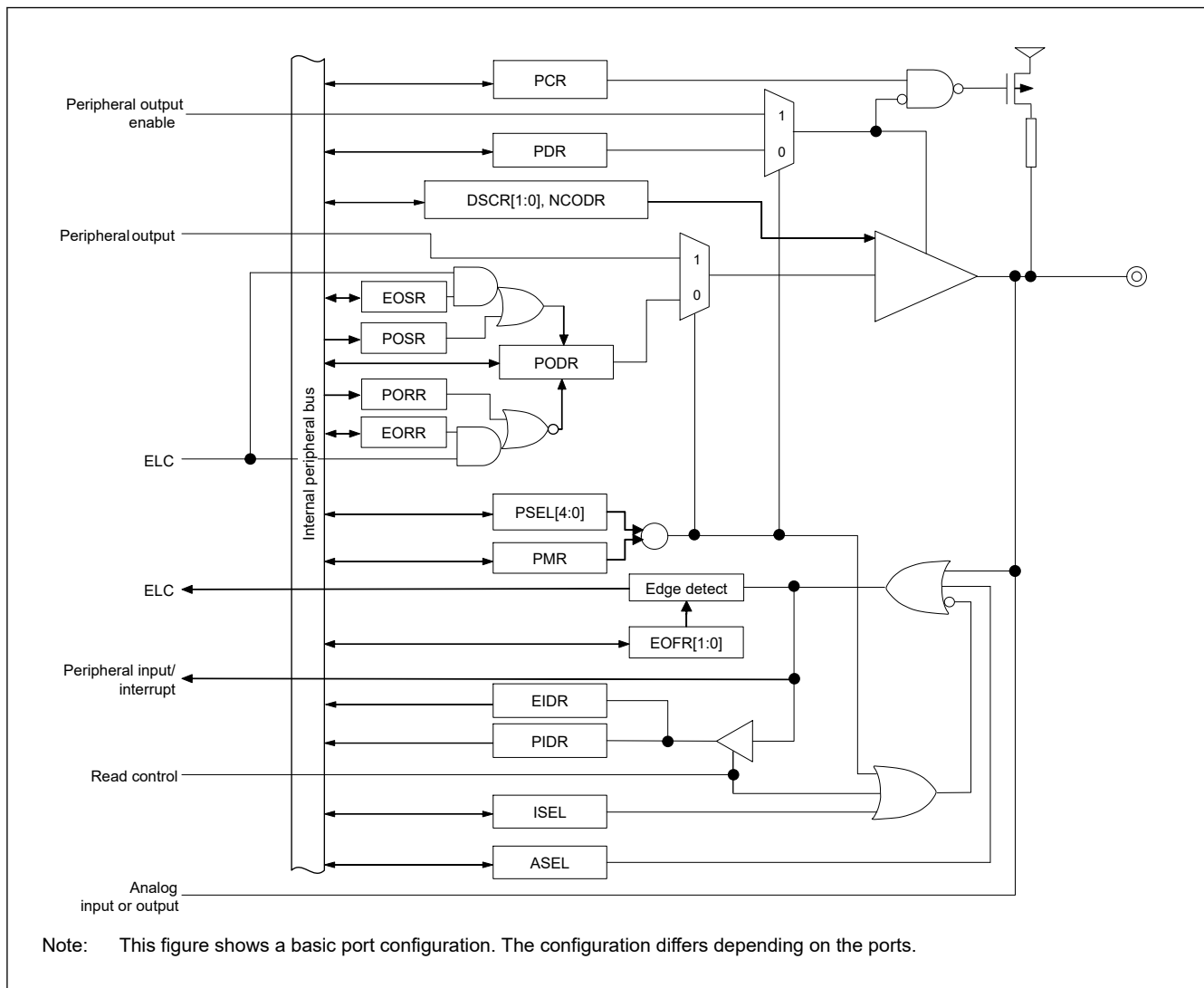


Figure 20.1 Connection diagram for I/O port registers

Table 20.1 I/O port specifications (1 of 2)

Port	Package		Package		Package	
	289, 303 pins	Number of Pins	224 pins	Number of Pins	176 pins	Number of Pins
PORT0	P000 to P015	16	P000 to P009, P014, P015	12	P000 to P009, P014, P015	12
PORT1	P100 to P115	16	P100 to P107, P112 to P115	12	P100 to P107, P112 to P115	12
PORT2	P200 to P215	16	P200 to P206, P208 to P215	15	P200, P201, P206, P208 to P215	11

Table 20.1 I/O port specifications (2 of 2)

Port	Package		Package		Package	
	289, 303 pins	Number of Pins	224 pins	Number of Pins	176 pins	Number of Pins
PORT3	P300 to P315	16	P300 to P302, P304 to P315	15	P300 to P302, P304 to P308, P310 to P312	11
PORT4	P400 to P415	16	P400 to P411, P414, P415	14	P400 to P410, P414, P415	13
PORT5	P500 to P515	16	P503 to P512	10	P503 to P512	10
PORT6	P600 to P615* ¹	16	P600 to P604, P608 to P615	13	P600, P601, P608 to P615	10
PORT7	P700 to P715	16	P700 to P714	15	P700 to P705, P707 to P709	9
PORT8	P800 to P815	16	P800 to P804, P808, P809, P813 to P815	10	P800 to P804, P808, P809, P813 to P815	10
PORT9	P900 to P915	16	P902, P905 to P909	6	P905 to P909	5
PORTA	PA00 to PA15	16	PA00 to PA15	16	PA00 to PA15	16
PORTB	PB00 to PB14	15	PB00 to PB04, PB08 to PB14	12	PB00 to PB04, PB08 to PB14	12
PORTC	PC00 to PC15* ¹	16	PC11 to PC15	5	PC11 to PC15	5
PORTD	PD00 to PD07	8	PD00	1	PD00	1

Note 1. For BGA303 package, P603 to P607 and PC00 to PC07 are connected to the SIP Flash memory inside the package.

Table 20.2 I/O port functions (1 of 2)

Port	Port name	Input pull-up	Open-drain output	Drive capability switching	5Vtolerant	I/O	Power supply
PORT0	P000 to P015	✓	✓	Low	—	Input / Output	AVCC0
PORT1	P100 to P107	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC2
	P108 to P115	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC
PORT2	P200	✓	—	—	—	Input	VCC
	P201	✓	✓	Low	—	Input / Output	VCC
	P202, P203	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC2
	P204, P205	✓	✓	Low, middle, high, high speed high drive	✓	Input / Output	VCC2
	P206 to P213	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC
	P214, P215	✓	—	—	—	Input	VBATT_R
PORT3	P300 to P302, P304 to P312	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC
	P303	✓	✓	Low, middle, high, high speed high drive	✓	Input / Output	VCC
	P313 to P315	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC2
PORT4	P400 to P406, P414, P415	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC
	P407 to P413	✓	✓	Low, middle, high, high speed high drive	✓	Input / Output	VCC

Table 20.2 I/O port functions (2 of 2)

Port	Port name	Input pull-up	Open-drain output	Drive capability switching	5Vtolerant	I/O	Power supply
PORT5	P500 to P502	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC2
	P503 to P510, P513	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC
	P511, P512, P514, P515	✓	✓	Low, middle, high, high speed high drive	✓	Input / Output	VCC
PORT6	P600 to P607	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC2
	P608 to P615	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC
PORT7	P700 to P707	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC
	P708 to P715	✓	✓	Low, middle, high, high speed high drive	✓	Input / Output	VCC
PORT8	P800 to P804, P808 to P812	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC2
	P805 to P807, P813	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC
	P814, P815	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC_USB
PORT9	P900, P901	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC2
	P902 to P915	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC
PORTA	PA00 to PA15	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC
PORTB	PB00 to PB14	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC
PORTC	PC00 to PC10	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC2
	PC11 to PC15	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC
PORTD	PD00 to PD07	✓	✓	Low, middle, high, high speed high drive	—	Input / Output	VCC

Note: ✓: Available
 —: Setting prohibited

20.2 Register Descriptions

20.2.1 PCNTR1/PODR/PDR : Port Control Register 1

Base address: $\text{PORTm} = 0x4040_0000 + 0x0020 \times m$ ($m = 0$ to 9, A, B, C, D)
 $\text{PORTm_NS} = 0x5040_0000 + 0x0020 \times m$ ($m = 0$ to 9, A, B, C, D)

Offset address: 0x000 (PCNTR1/PDR)
 0x002 (PODR)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	PODR 15	PODR 14	PODR 13	PODR 12	PODR 11	PODR 10	PODR 09	PODR 08	PODR 07	PODR 06	PODR 05	PODR 04	PODR 03	PODR 02	PODR 01	PODR 00
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	PDR1 5	PDR1 4	PDR1 3	PDR1 2	PDR11	PDR1 0	PDR0 9	PDR0 8	PDR0 7	PDR0 6	PDR0 5	PDR0 4	PDR0 3	PDR0 2	PDR0 1	PDR0 0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	PDR15 to PDR00	Pmn Direction 0: Input (functions as an input pin) 1: Output (functions as an output pin)	R/W
31:16	PODR15 to PODR00	Pmn Output Data 0: Low output 1: High output	R/W

Note: S-TYPE-4, P-TYPE-5

Note: $m = 0$ to 9, A, B, C, D, $n = 00$ to 15

The Port Control Register 1 (PCNTR1/PODR/PDR) is a 32-bit or 16-bit read/write register that controls port direction and port output data. The PCNTR1 specifies the port direction and output data, and is accessed in 32-bit units. The PDRn (bits [15:0] in PCNTR1) and PODRn (bits [31:16] in PCNTR1) respectively, are accessed in 16-bit units.

PDRn bits (Pmn Direction)

The PDRn bits select the input or output direction for individual pins on the associated port when the pins are configured as general I/O pins. Each pin on port m is associated with a PORTm.PCNTR1.PDRn bit. The I/O direction can be specified in 1-bit unit. Bits associated with non-existent pins are reserved. Reserved bits are read as 0. The write value should be 0. In the case of input only ports, PDRn bits are reserved. See [Table 20.2](#). The PDRn bit in the PORTm.PCNTR1 register serves the same function as the PDR bit in the PFS.PmnPFS register.

PODRn bits (Pmn Output Data)

The PODRn bits hold data to be output from the general I/O pins. Bits of non-existent port m are reserved. Reserved bits are read as 0. The write value should be 0. In the case of input only ports, PODRn bits are reserved. See [Table 20.2](#). The PODRn bit in the PORTm.PCNTR1 register serves the same function as the PODR bit in the PFS.PmnPFS register.

20.2.2 PCNTR2/EIDR/PIDR : Port Control Register 2

Base address: PORTm = 0x4040_0000 + 0x0020 × m (m = 0 to 9, A, B, C, D)
 PORTm_NS = 0x5040_0000 + 0x0020 × m (m = 0 to 9, A, B, C, D)

Offset address: 0x004 (PCNTR2/PIDR)
 0x006 (EIDR)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	EIDR1 5	EIDR1 4	EIDR1 3	EIDR1 2	EIDR1 1	EIDR1 0	EIDR0 9	EIDR0 8	EIDR0 7	EIDR0 6	EIDR0 5	EIDR0 4	EIDR0 3	EIDR0 2	EIDR0 1	EIDR0 0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	PIDR1 5	PIDR1 4	PIDR1 3	PIDR1 2	PIDR1 1	PIDR1 0	PIDR0 9	PIDR0 8	PIDR0 7	PIDR0 6	PIDR0 5	PIDR0 4	PIDR0 3	PIDR0 2	PIDR0 1	PIDR0 0
Value after reset:	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

Bit	Symbol	Function	R/W
15:0	PIDR15 to PIDR00	Pmn State 0: Low level 1: High level	R
31:16	EIDR15 to EIDR00 ^{*2}	Port Event Input Data ^{*1} When an ELC_PORTx signal occurs 0: Low input 1: High input	R

Note: S-TYPE-4, P-TYPE-5

Note: m = 0 to 9, A, B, C, D, n = 00 to 15

Note 1. x = 1, 2, 3 or 4 for EIDR only

Note 2. Supported by ports 1, 2, 3 or 4.

The Port Control Register 2 (PCNTR2/EIDR/PIDR) allows read access to the Pmn state and the port event input data using 32-bit or 16-bit access.

The PCNTR2 specifies the Pmn state and the port event input data, and is accessed in 32-bit units.

The PIDRn (bits [15:0] in PCNTR2) and EIDRn (bits [31:16] in PCNTR2) respectively, are accessed in 16-bit units. Bits associated with non-existent pins are reserved. Reserved bits are read as undefined.

PIDRn bits (Pmn State)

The PIDRn bits reflect the individual pin states of the port, regardless of the values set in PmnPFS.PMR and PORTm.PCNTR1.PDRn. The PIDRn bit in the PORTm.PCNTR2 register serves the same function as the PIDR bit in the PFS.PmnPFS register.

A pin state cannot be reflected in PIDRn when one of the following functions is enabled:

- RTC Time Capture input and tamper detection (RTCIC)
- Analog function (ASEL = 1)
- I3C bus interface (I3C_SCL0, I3C_SDA0)
- JTAG/SWD interface (TCK/SWCLK, TMS/SWDIO, TDI, TDO)
- Trace interface (TCLK, TDATA_n (n = 0 to 3), SWO)

EIDRn bits (Port Event Input Data)

The EIDRn bits latch a pin state when an ELC_PORTx signal occurs. Pin states can only be input to EIDRn when PmnPFS.PMR and PORTm.PCNTR1.PDRn are 0. When the PmnPFS.ASEL bit is set to 1, the associated pin state is not reflected in EIDRn.

20.2.3 PCNTR3/PORR/POSR : Port Control Register 3

Base address: $\text{PORTm} = 0x4040_0000 + 0x0020 \times m$ ($m = 0$ to 9, A, B, C, D)
 $\text{PORTm_NS} = 0x5040_0000 + 0x0020 \times m$ ($m = 0$ to 9, A, B, C, D)

Offset address: 0x008 (PCNTR3/POSR)
 0x00A (PORR)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	PORR 15	PORR 14	PORR 13	PORR 12	PORR 11	PORR 10	PORR 09	PORR 08	PORR 07	PORR 06	PORR 05	PORR 04	PORR 03	PORR 02	PORR 01	PORR 00
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	POSR 15	POSR 14	POSR 13	POSR 12	POSR 11	POSR 10	POSR 09	POSR 08	POSR 07	POSR 06	POSR 05	POSR 04	POSR 03	POSR 02	POSR 01	POSR 00
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	POSR15 to POSR00	Pmn Output Set 0: No effect on output 1: High output	W
31:16	PORR15 to PORR00	Pmn Output Reset 0: No effect on output 1: Low output	W

Note: S-TYPE-4, P-TYPE-5

Note: $m = 0$ to 9, A, B, C, D, $n = 00$ to 15

The Port Control Register 3 (PCNTR3/PORR/POSR) is a 32-bit or 16-bit write register that controls the setting or resetting of the port output data.

The PCNTR3 controls the setting or resetting of the port output data, and is accessed in 32-bit units.

The POSRn (bits [15:0] in PCNTR3) and the PORRn (bits [31:16] in PCNTR3) respectively, are accessed in 16-bit units.

POSRn bits (Pmn Output Set)

POSR changes PODR when set by a software write. For example, for P100, when $\text{PORT1.PCNTR3.POSR00} = 1$, $\text{PORT1.PCNTR1.PODR00}$ outputs 1. Bits associated with non-existent pins are reserved. The write value should always be 0. In the case of input only ports, POSRn bits are reserved. See [Table 20.2](#).

PORRn bits (Pmn Output Reset)

PORR changes PODR when reset by a software write. For example, for P100, when $\text{PORT1.PCNTR3.PORR00} = 1$, $\text{PORT1.PCNTR1.PODR00}$ outputs 0. Bits associated with non-existent pins are reserved. The write value should always be 0. In the case of input only ports, PORRn bits are reserved. See [Table 20.2](#).

Note: When EORRn or EOSRn is set, writing is prohibited to PODRn, PORRn, and POSRn.

Note: PORRn and POSRn should not be set at the same time.

20.2.4 PCNTR4/EORR/EOSR : Port Control Register 4

Base address: PORTm = 0x4040_0000 + 0x0020 × m (m = 1 to 4)
 PORTm_NS = 0x5040_0000 + 0x0020 × m (m = 1 to 4)

Offset address: 0x00C (PCNTR4/EOSR)
 0x00E (EORR)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	EORR 15	EORR 14	EORR 13	EORR 12	EORR 11	EORR 10	EORR 09	EORR 08	EORR 07	EORR 06	EORR 05	EORR 04	EORR 03	EORR 02	EORR 01	EORR 00
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	EOSR 15	EOSR 14	EOSR 13	EOSR 12	EOSR 11	EOSR 10	EOSR 09	EOSR 08	EOSR 07	EOSR 06	EOSR 05	EOSR 04	EOSR 03	EOSR 02	EOSR 01	EOSR 00
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	EOSR15 to EOSR00	Pmn Event Output Set When an ELC_PORTx signal occurs 0: No effect on output 1: High output	R/W
31:16	EORR15 to EORR0	Pmn Event Output Reset When an ELC_PORTx signal occurs 0: No effect on output 1: Low output	R/W

Note: S-TYPE-4, P-TYPE-5

Note: m = 1 to 4, n = 00 to 15, x = 1 to 4

The Port Control Register 4 (PCNTR4/EORR/EOSR) is a 32-bit or 16-bit read/write register that controls the setting or resetting of the port output data by an event input from the ELC.

The PCNTR4 controls the setting or resetting of the port output data by an event input from the ELC, and is accessed in 32-bit units.

The EOSRn (bits [15:0] in PCNTR4) and EORRn (bits [31:16] in PCNTR4) respectively, are accessed in 16-bit units.

EOSRn bits (Pmn Event Output Set)

EOSR changes PODR when set because an ELC_PORTx signal occurs. For example, for P100 if PORT1.PCNTR4.EOSR00 is set to 1 when the ELC_PORTx occurs, PORT1.PCNTR1.PODR00 outputs 1. Bits associated with non-existent pins are reserved. The write value should always be 0. For input only ports, EOSRn bits are reserved. See [Table 20.2](#).

EORRn bits (Pmn Event Output Reset)

EORR changes PODR when reset because an ELC_PORTx signal occurs. For example, for P100 if PORT1.PCNTR4.EORR00 = 1 when the ELC_PORTx occurs, PORT1.PCNTR1.PODR00 outputs 0. Bits associated with non-existent pins are reserved. The write value should always be 0. For input only ports, EORRn bits are reserved. See [Table 20.2](#).

Note: When EORRn or EOSRn is set, writing is prohibited to PODRn, PORRn, and POSRn.

Note: EORRn and EOSRn should not be set at the same time.

20.2.5 PmnPFS/PmnPFS_HA/PmnPFS_BY : Port mn Pin Function Select Register (m = 0 to 9, A, B, C, D, n = 00 to 15)

Base address: PFS = 0x4040_0800
PFS_NS = 0x5040_0800

Offset address: 0x000 + 0x040 × m + 0x004 × n (PmnPFS/PmnPFS_HA/PmnPFS_BY)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	PSEL[4:0]				—	—	—	—	—	—	—	—	PMR
Value after reset:	0	0	0	0 ^{*1}	0	0	0 ^{*1}	0 ^{*1}	0	0	0	0	0	0	0	0 ^{*1}
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	ASEL	ISEL	EOFR[1:0]		DSCR[1:0]		—	—	—	NCOD R	—	PCR	—	PDR	PIDR	PODR
Value after reset:	0	0	0	0	0	0 ^{*1}	0	0	0	0	0	0 ^{*1}	0	0	x	0

Bit	Symbol	Function	R/W
0	PODR	Port Output Data 0: Low output 1: High output	R/W
1	PIDR	Pmn State 0: Low level 1: High level	R
2	PDR	Port Direction 0: Input (functions as an input pin) 1: Output (functions as an output pin)	R/W
3	—	This bit is read as 0. The write value should be 0.	R/W
4	PCR	Pull-up Control 0: Disable input pull-up 1: Enable input pull-up	R/W
5	—	This bit is read as 0. The write value should be 0.	R/W
6	NCODR	N-Channel Open-Drain Control 0: CMOS output 1: NMOS open-drain output	R/W
9:7	—	These bits are read as 0. The write value should be 0.	R/W
11:10	DSCR[1:0]	Port Drive Capability 0 0: Low drive 0 1: Middle drive 1 0: High speed high drive 1 1: High drive	R/W
13:12	EOFR[1:0]	Event on Falling/Event on Rising ^{*2} 0 0: Don't care 0 1: Detect rising edge 1 0: Detect falling edge 1 1: Detect both edges	R/W
14	ISEL	IRQ Input Enable 0: Not used as an IRQn input pin 1: Used as an IRQn input pin	R/W
15	ASEL	Analog Input Enable 0: Not used as an analog pin 1: Used as an analog pin	R/W
16	PMR	Port Mode Control 0: Used as a general I/O pin 1: Used as an I/O port for peripheral functions	R/W
23:17	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
28:24	PSEL[4:0]	Peripheral Select These bits select the peripheral function. For individual pin functions, see the associated tables in this chapter.	R/W
31:29	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

Note 1. The initial value of P201, P208, P209, P210, P211, P814 and P815 is not 0x0000_0000. P201 is 0x0000_0010, P208 is 0x0001_0010, P209 is 0x0001_0400, P210 is 0x0001_0410, P211 is 0x0001_0010, P814 is 0x1301_0000 and P815 is 0x1301_0000.

Note 2. Supported by PORTn (n = 1 to 4).

Port mn Pin Function Select Register (PmnPFS/PmnPFS_HA/PmnPFS_BY) is a 32-bit, 16-bit, or 8-bit read/write control register that selects the port mn pin function, and is accessed in 32-bit units. PmnPFS_HA (PmnPFS [15:0] bits) is accessed in 16-bit units. PmnPFS_BY (PmnPFS[7:0] bits) is accessed in 8-bit units.

The available Port mn pin depends on the product. For details, see [Table 20.1](#)

PODR bit (Port Output Data), PIDR bit (Port State), PDR bit (Port Direction)

The PDR, PIDR, and PODR bits serve the same function as the PCNTR. When these bits are read, the PCNTR value is read.

PCR bit (Pull-up Control)

The PCR bit enables or disables an input pull-up resistor on the individual port pins. When a pin is in the input state with the associated bit in PmnPFS.PCR set to 1, the pull-up resistor connected to the pin is enabled. When a pin is set as a general port output pin, or a peripheral function output pin, the pull-up resistor for the pin is disabled regardless of the PCR setting. The pull-up resistor is also disabled in the reset state. Bits associated with non-existent pins are reserved. Reserved bits are read as 0. The write value should be 0.

NCODR bit (N-Channel Open-Drain Control)

The NCODR bit specifies the output type for the port pins. Bits associated with non-existent pins are reserved. Reserved bits are read as 0. The write value should be 0.

DSCR[1:0] bits (Port Drive Capability)

The DSCR[1:0] bits switch the drive capacity of the port. If the drive capacity of a pin is fixed, the associated bit is a read/write bit, but the drive capacity cannot be changed. Bits associated with non-existent pins are reserved. Reserved bits are read as 0. The write value should be 0.

EOFR[1:0] bits (Event on Falling/Event on Rising)

The EOFR[1:0] bits select the edge detection method for the port group input signal. These bits support rising, falling, or both edge detections. When the EOFR[1:0] bits are set to 01b, 10b, or 11b, the input enable of the I/O cell is asserted. Following that, the event pulse is input from the external pin, and the GPIO outputs the event pulse to the ELC. Bits associated with non-existent pins are reserved. Reserved bits are read as 0. The write value should be 0.

ISEL bit (IRQ Input Enable)

The ISEL bit specifies IRQ input pins. This setting can be used in combination with the peripheral functions, although an IRQn (external pin interrupt) of the same number must only be enabled for one pin. The ISEL bit for an unspecified IRQn is reserved.

ASEL bit (Analog Input Enable)

The ASEL bit specifies analog pins. When a pin is set as an analog pin by this bit:

1. Specify it as a general I/O port in the Port Mode Control bit (PmnPFS.PMR).
2. Disable the pull-up resistor in the Pull-up Control bit (PmnPFS.PCR).
3. Specify the input in the Port Direction bit (PmnPFS.PDR). The pin state cannot be read at this point. The PmnPFS register is protected by the Write-Protect Register for Non-secure (PWPR_NS). Release write-protect before modifying the register.

The ASEL bit for an unspecified analog I/O pin is reserved.

PMR bit (Port Mode Control)

The PMR bit specifies the port pin function. Bits associated with non-existent pins are reserved. The write value should be 0.

PSEL[4:0] bits (Peripheral Select)

The PSEL[4:0] bits assign the peripheral function.

20.2.6 PWPR_NS : Write-Protect Register for Non-secure

Base address: PFS_NS = 0x5040_0800

Offset address: 0x50C

Bit position:	7	6	5	4	3	2	1	0
Bit field:	B0WI	PFSWE	—	—	—	—	—	—
Value after reset:	1	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
5:0	—	These bits are read as 0. The write value should be 0.	R/W
6	PFSWE	PmnPFS Register Write Enable 0: Writing to the PmnPFS register is disabled 1: Writing to the PmnPFS register is enabled	R/W
7	B0WI	PFSWE Bit Write Disable 0: Writing to the PFSWE bit is enabled 1: Writing to the PFSWE bit is disabled	R/W

Note: S-TYPE-7, P-TYPE-2

Note: The access to this register is not controlled by any security attribution register.

PFSWE bit (PmnPFS Register Write Enable)

Writing to the PmnPFS register is enabled only when the PFSWE bit is set to 1. You must first write 0 to the B0WI bit before setting PFSWE to 1.

B0WI bit (PFSWE Bit Write Disable)

Writing to the PFSWE bit is enabled only when the B0WI bit is set to 0.

20.2.7 PFENET : Ethernet Control Register

Base address: PFS = 0x4040_0800
PFS_NS = 0x5040_0800

Offset address: 0x500

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	ETHIOMD1	ETHIOMD0	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	—	These bits are read as 0. The write value should be 0.	R/W
4	ETHIOMD0	Ethernet I/O Mode Setting ch0 0: GMII mode (ESWM channel 0) 1: MII mode (ESWM channel 0)	R/W
5	ETHIOMD1	Ethernet I/O Mode Setting ch1 0: GMII mode (ESWM channel 1) 1: MII mode (ESWM channel 1)	R/W
7:6	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: The access to the ETHIOMD0/1 bit is controlled by the PSARC.PSARC30 bit and PPARC.PPARC30 bit.

ETHIOMD0/ETHIOMD1 bit (Ethernet I/O Mode Setting ch0 and Ethernet I/O Mode Setting ch1)

When the transmit data output ports of ESWM are used as GMII/RGMII ports, the drive capability can be set by the ETHIOMDn bit and DSCR[1:0] bit. The details of the setting are given by the following table:

Interface type	PSEL[4:0]	ETHIOMDn	VCC voltage [V]	DSCR [1:0]	Drive capability
MII only	0x16	1b	3.6-2.7	01b	Middle drive
MII/GMII		0b	3.6-3.0	01b	RGMII/3.3 V
			2.7-2.3	11b	RGMII/2.5 V
RMII	0x17	—	3.6-2.7	11b	High drive
RGMII	0x18		3.6-3.0	01b	RGMII/3.3 V
			2.7-2.3	11b	RGMII/2.5 V

If the settings of the ETHIOMDn bit and DSCR[1:0] bit other than those in the above table are performed, the electrical characteristics cannot be guaranteed.

20.2.8 PWPR_S : Write-Protect Register for Secure

Base address: PFS = 0x4040_0800

Offset address: 0x514

Bit position:	7	6	5	4	3	2	1	0
Bit field:	B0WI	PFSWE	—	—	—	—	—	—

Value after reset: 1 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
5:0	—	These bits are read as 0. The write value should be 0.	R/W
6	PFSWE	PmnPFS Register Write Enable 0: Disable writes to the PmnPFS register 1: Enable writes to the PmnPFS register	R/W
7	B0WI	PFSWE Bit Write Disable 0: Enable writes the PFSWE bit 1: Disable writes to the PFSWE bit	R/W

Note: S-TYPE-6, P-TYPE-2

PFSWE bit (PmnPFS Register Write Enable)

Writing to the PmnPFS register of the IO port pin set as secure by the PmSAR register is enabled only when the PFSWE bit is set to 1. You must first write 0 to the B0WI bit before setting PFSWE to 1.

B0WI bit (PFSWE Bit Write Disable)

Writing to the PFSWE bit is enabled only when the B0WI bit is set to 0.

20.2.9 PmSAR : Port Security Attribution Register (m = 0 to 9, A, B, C, D)

Base address: PFS = 0x4040_0800
PFS_NS = 0x5040_0800

Offset address: 0x530 + 0x004 × m

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	PMNSA[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	PMNSA[15:0]	Pmn Security Attribution Target I/O port pin : Pmn 0: Secure 1: Non-secure	R/W

Note: S-TYPE-1, P-TYPE-1

Note: This register is write-protected by PRCR register.

Note: m = 0 to 9, A, B, C, D, n = 00 to 15

Port Security Attribution Register is a 16-bit register that setting the Security Attribution of each port, the registers are accessed only in 16-bit units.

PMNSA[15:0] bits (Pmn Security Attribution)

The PmnSA bit specifies the Security Attribution of Pmn.

20.2.10 LVOCR : Low Voltage Operation Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xAB0

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	LVO1E	LVO0E
Value after reset:	0	0	0	0	0	0	1	1

Bit	Symbol	Function	R/W
0	LVO0E	Low Voltage Operation 0 Enable 0: Disable 1: Enable	R/W
1	LVO1E	Low Voltage Operation 1 Enable 0: Disable 1: Enable	R/W
7:2	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

Low voltage operation control register controls the IO when VCC/VCC2 is lower than 2.7V.

LVO0E bit (Low Voltage Operation 0 Enable)

When VCC is lower than 2.7V, set the LVO0E bit to 1.

LVO1E bit (Low Voltage Operation 1 Enable)

When VCC2 is lower than 2.7V, set the LVO1E bit to 1.

20.3 Operation

20.3.1 General I/O Ports

All pins except P208 to P211, P814 and P815 operate as general I/O ports after reset. General I/O ports are organized as 16 bits per port and can be accessed by port with the Port Control Registers (PCNTRn, where n = 1 to 4), or by individual pins with the Port mn Pin Function Select register. For details on these registers, see [section 20.2. Register Descriptions](#).

Each port has the following bits:

- Port Security Attribution register (PmSAR) (m = 0 to 9, A, B, C, D), which indicates the security attribution.
- Port Direction bit (PDRn), which selects input or output direction
- Port Output Data bit (PODRn), which holds data for output
- Port Input Data bit (PIDRn), which indicates the pin states
- Event Input Data bit (EIDRn), which indicates the pin state when an ELC_PORTn (n = 1, 2, 3 or 4) signal occurs
- Port Output Set bit (POSRn), which indicates the output value when a software write occurs
- Port Output Reset bit (PORRn), which indicates the output value when a software write occurs
- Event Output Set bit (EOSRn), which indicates the output value when an ELC_PORTn (n = 1, 2, 3 or 4) signal occurs
- Event Output Reset bit (EORRn), which indicates the output value when an ELC_PORTn (n = 1, 2, 3 or 4) signal occurs.

20.3.2 Port Function Select

The following port functions are available for configuring each pin:

- Security function: Security attribution for each pins
- I/O configuration: CMOS output or NMOS open-drain output, pull-up control, and drive capacity
- General I/O port: Port direction, output data setting, and read input data
- Alternate function: Configured function mapping to the pin.

Each pin is associated with a Port mn Pin Function Select register (PmnPFS), which includes the associated PODR, PIDR, and PDR bits. In addition, the PmnPFS register includes the following:

- PCR: Pull-up resistor control bit that turns the input pull-up MOS on or off
- NCODR: N-channel open-drain control bit that selects the output type for each pin
- DSCR[1:0]: Drive capacity control bit that selects the drive capacity
- EOFR[1:0]: For selecting the edge of the event that input from the port group
- ISEL: IRQ input enable bit to specify an IRQ input pin
- ASEL: Analog input enable bit to specify an analog pin
- PMR: Port mode bit to specify the pin function of each port
- PSEL[4:0]: Port function select bits to select the associated peripheral function.

These configurations can be made by a single-register access to the Port mn Pin Function Select register. For details, see [section 20.2.5. PmnPFS/PmnPFS_HA/PmnPFS_BY : Port mn Pin Function Select Register \(m = 0 to 9, A, B, C, D, n = 00 to 15\)](#).

20.3.3 Port Group Function for ELC

In the MCU, PORT1 to PORT4 are assigned for the ELC port group function.

20.3.3.1 Behavior When ELC_PORTn (n = 1, 2, 3 or 4) is Input from ELC

The MCU supports the two functions described in this section when an ELC_PORTn (n = 1, 2, 3 or 4) signal comes from the ELC.

(1) Input to EIDR

For the GPI function ($PDR = 0$ and $PMR = 0$ in the $PmnPFS$ register), when an ELC_PORTn ($n = 1, 2, 3$ or 4) signal comes from the ELC, the input enable of the I/O cell is asserted, and data from the external pins is read into the EIDR bit. See [Figure 20.2](#)

For the GPO function ($PDR = 1$) or the peripheral mode ($PMR = 1$), 0 is input into the EIDR bit from the external pins.

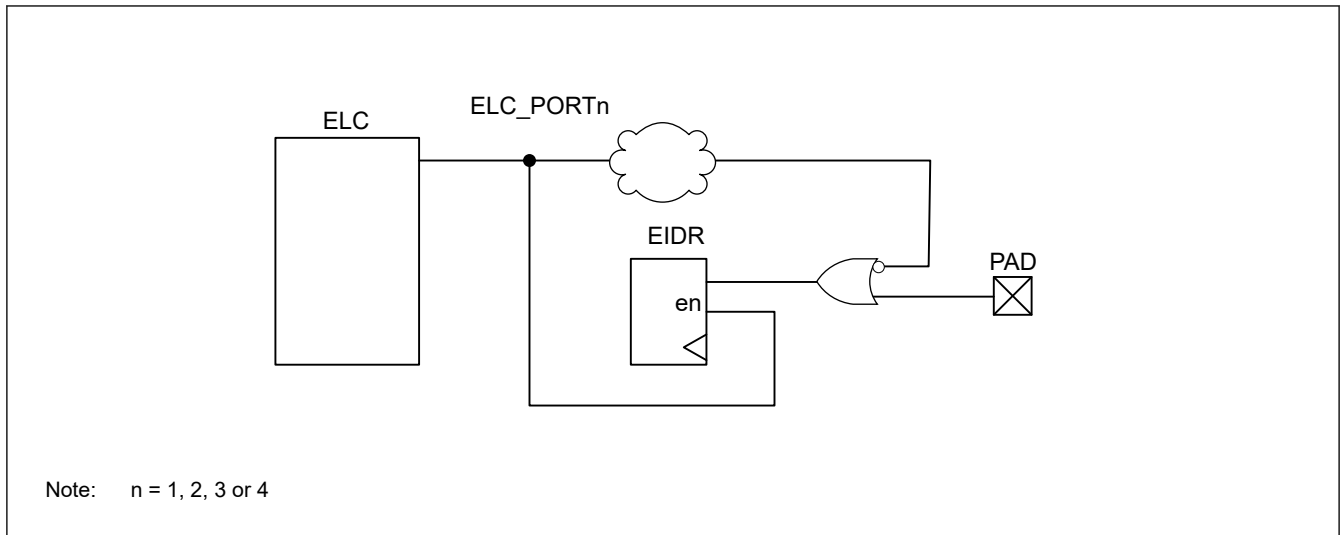


Figure 20.2 Event ports input data

(2) Output from PODR by EOSR and EORR

When an ELC_PORTn ($n = 1, 2, 3$ or 4) signal occurs, the data is output from the PODR to the external pin based on the settings in the EOSR and EORR registers.

- If EOSR is set to 1, when an ELC_PORTn ($n = 1, 2, 3$ or 4) signal occurs, the PODR register outputs 1 to the external pin. Otherwise, when EOSR = 0, the PODR value is retained.
- If EORR is set to 1, when ELC_PORTn ($n = 1, 2, 3$ or 4) signal occurs, the PODR register outputs 0 to the external pin. Otherwise, when EORR = 0, the PODR value is retained.

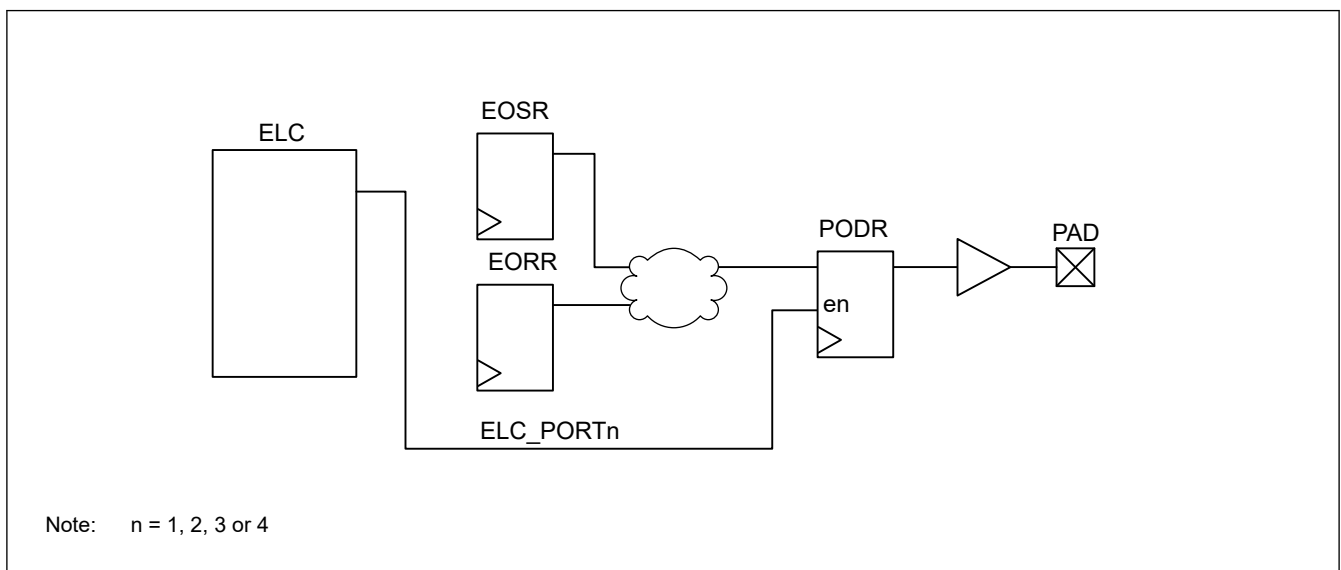


Figure 20.3 Event ports output data

20.3.3.2 Behavior When an Event Pulse is Output to ELC

To output the event pulse from the external pins to the ELC, set the EOFR[1:0] bits in the PmnPFS register. For details, see [section 20.2.5. PmnPFS/PmnPFS_HA/PmnPFS_BY : Port mn Pin Function Select Register \(m = 0 to 9, A, B, C, D, n = 00 to 15\)](#). When the EOFR[1:0] bits are set, the input enable of the I/O cell is asserted.

Data from the external pin is the input. For example, for Port 1, when the data is input from P100 to P115, the data of those 16 pins is organized by OR logic. This data is formed into a one-shot pulse that goes to the ELC. The operation of Port n (n = 2 to 4) is also the same as Port 1. See [Figure 20.4](#).

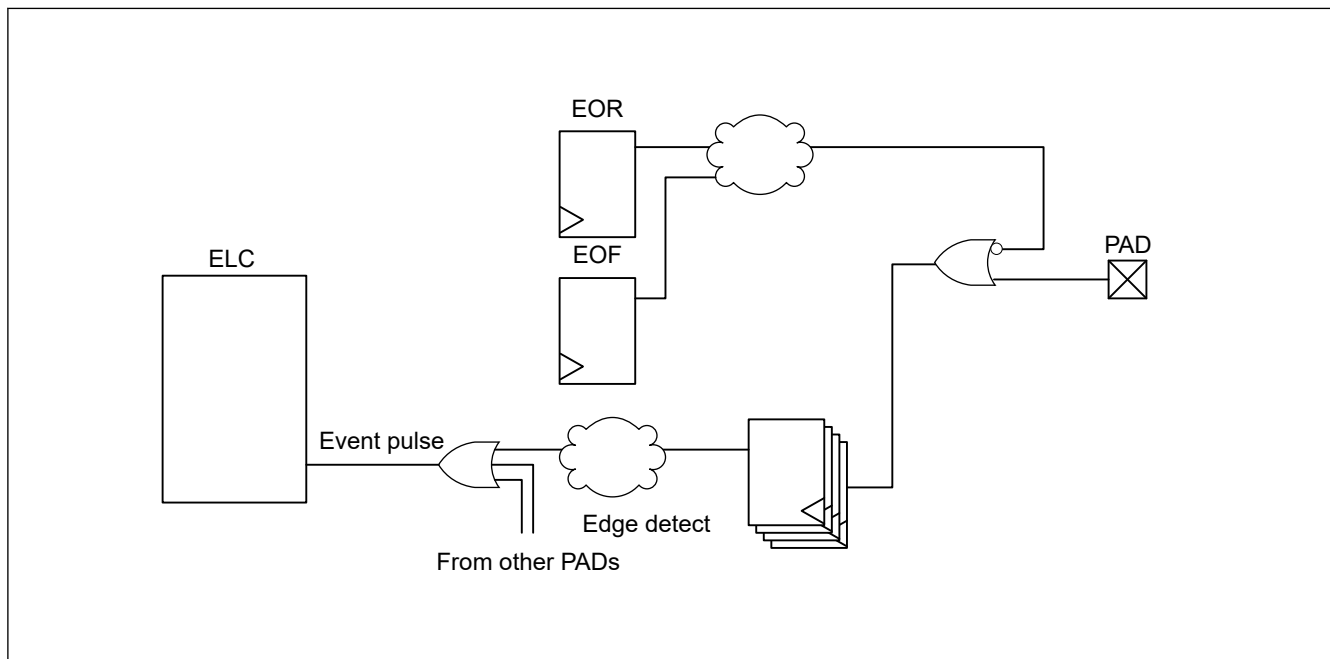


Figure 20.4 Generation of event pulse

20.4 Handling of Unused Pins

[Table 20.3](#) shows how to handle unused pins.

Table 20.3 Handling of unused pins (1 of 2)

Pin name	Description
MD	Use as a mode selection pin
RES	Connect to VCC through a resistor
USB_DP, USB_DM	- Preconditions: - VCC_USB: Connect to VCC - VSS_USB: Connect to VSS - Set the module-stop state for USBFS (MSTPCRB.MSTPB11 = 1) - Processing details: - When both P814PFS.PMR and P815PFS.PMR bits are set to 1, keep these pins open. - When P814PFS.PMR or P815PFS.PMR bit is set to 0, configure it in the same way as P814, P815.
P200/NMI	Connect to VCC through a resistor
EXTAL	When the main clock oscillator is not used, set the MOSCCR.MOSTP bit to 1 (general port P212). When this pin is not used as port P212, configure it in the same way as Other ports.
XTAL	When the main clock oscillator is not used, set the MOSCCR.MOSTP bit to 1 (general port P213). When the external clock is input to the XTAL pin, the XTAL pin functions as P213. When this pin is not used as port P213, configure it in the same way as Other ports.
XCIN/EXCIN	When the sub clock oscillator is not used, set the SOSCCR.SOSTP bit to 1 and set the SOMCR.SOSEL bit to 0 (general port P215). When this pin is not used as port P215, configure it in the same way as P214, P215.

Table 20.3 Handling of unused pins (2 of 2)

Pin name	Description
XCOUT	When the sub clock oscillator is not used, set the SOSCCR.SOSTP bit to 1 (general port P214). When the external clock is input to the EXCIN pin, the XCOUT pin functions as P214. When this pin is not used as port P214, configure it in the same way as P214, P215.
P000 to P015	- If the direction is set to input (PCNTR1.PDRn = 0), connect the associated pin to AVCC0 through a resistor or to AVSS0 through a resistor ^{*1} - If the direction is set to output (PCNTR1.PDRn = 1), keep pin open ^{*1}
P100 to P107, P202 to P205, P313 to P315, P500 to P502, P600 to P607, P800 to P804, P808 to P812, P900, P901, PC00 to PC10	- If the direction is set to input (PCNTR1.PDRn = 0), connect the associated pin to VCC2 through a resistor or to VSS through a resistor ^{*1} - If the direction is set to output (PCNTR1.PDRn = 1), keep pin open ^{*1}
P214, P215	Connect to VSS through a resistor
P814, P815	- If the direction is set to input (PCNTR1.PDRn = 0), connect the associated pin to VCC_USB through a resistor or to VSS through a resistor ^{*1} - If the direction is set to output (PCNTR1.PDRn = 1), keep pin open ^{*1}
Other ports	- If the direction is set to input (PCNTR1.PDRn = 0), connect the associated pin to VCC through a resistor or to VSS through a resistor ^{*1 *2} - If the direction is set to output (PCNTR1.PDRn = 1), keep pin open ^{*1 *3}
VREFH0, VREFH	Connect to AVCC0
VREFL0, VREFL	Connect to AVSS0
VLO	Keep pin open ^{*4}
VBATT	Connect to VCC

Note 1. Clear the PmnPFS.PMR, PmnPFS.ISEL, PmnPFS.PCR, and PmnPFS.ASEL bits to 0.

Note 2. P208, P210 and P211 are recommended for pull up VCC through a resistor, because these pins are input pull-up enabled from the initial value (PmnPFS.PCR = 1).

Note 3. P209 is recommended for setting the direction to output (PCNTR1.PDRn = 1), because this pin is output from the initial value.

Note 4. For more information, see [section 59, Internal Voltage Regulator](#)

20.5 Usage Notes

20.5.1 Procedure for Specifying the Pin Functions

To specify the I/O pin functions:

1. Clear the B0WI bit in the PWPR_NS register. This enables writing to the PFSWE bit in the PWPR_NS register.^{*1}
2. Set 1 to the PFSWE bit in the PWPR_NS register. This enables writing to the PmnPFS register.^{*1}
3. Clear the Port Mode Control bit in the PMR to 0 for the target pin to select the general I/O port.
4. Specify the I/O function for the pin through the PSEL[4:0] bits settings in the PmnPFS register.
5. Set the PMR bit to 1 as required to switch to the selected I/O function for the pin.
6. Clear the PFSWE bit in the PWPR_NS register. This disables writing to the PmnPFS register.^{*1}
7. Set 1 to the B0WI bit in the PWPR_NS register. This disables writing to the PFSWE bit in the PWPR_NS register.^{*1}

Note 1. When the security attribution of Pmn is set to 0, set the PWPR_S register to write to the PmnPFS register.

20.5.2 Procedure for Using Port Group Input

To use the port group input (port n (n = 1 to 4)):

1. Set the ELSRx.ELS[8:0] bits to all 0 to ignore unexpected pulses. For more information, see [section 19, Event Link Controller \(ELC\)](#).

2. Set the EOFR[1:0] bits of the PmnPFS register to specify the rising, falling, or both edge detections.
3. Execute a dummy read or wait for a short time, for example 100 ns. Ignoring of unexpected pulses depends on the initial value of the external pin.
4. Set the ELSRx.ELS[8:0] bits to enable the event signals.

20.5.3 Port Output Data Register (PODR) Summary

This register outputs data as follows:

1. Outputs 0 if PCNTR4.EORR is set to 1 when ELC_PORTn (n = 1, 2, 3 or 4) signal occurs.
2. Outputs 1 if PCNTR4.EOSR is set to 1 when ELC_PORTn (n = 1, 2, 3 or 4) signal occurs.
3. Outputs 0 if PCNTR3.PORR is set to 1.
4. Outputs 1 if PCNTR3.POSR is set to 1.
5. Outputs 0 or 1 because PCNTR1.PODRn is set.
6. Outputs 0 or 1 because PmnPFS.PODRn is set.

Numbers in this list correspond to the priority for writing to the PODRn. For example, if 1. and 3. from the list occur at the same time, the higher priority event 1. is executed.

20.5.4 Notes on Using Analog Functions

To use an analog function, set the Port Mode Control bit (PMR) and the Port Direction bit (PDRn) to 0 so that the pin acts as a general input port. Next, set the Analog Input Enable bit (ASEL) in the Port mn Pin Function Select Register (PmnPFS.ASEL) to 1.

20.5.5 I/O Buffer Specification

The P402, P403, and P404 can be used as the RTC input, tamper detection and other peripheral functions. [Table 20.4](#) lists the P402, P403, P404 specifications.

Table 20.4 P402, P403, P404 specifications

I/O port	RTC and tamper detection			Other peripheral	
	RTC and tamper detection input enable register	RTC	Tamper detection	Other peripheral enable register	ADC16H, DSMIF, GPT, CAC, ESWM, ESC, CANFD, SCI, SDHI and interrupt
P402	VBTICTLR.VCH0INEN	RTCIC0	RTCIC0	P402PFS.PSEL and PMR	For details, see section 20.6. Peripheral Select Settings for Each Product .
P403	VBTICTLR.VCH1INEN	RTCIC1	RTCIC1	P403PFS.PSEL and PMR	
P404	VBTICTLR.VCH2INEN	RTCIC2	RTCIC2	P404PFS.PSEL and PMR	

These RTC inputs and tamper detection are controlled by the VBTICTLR register and this register has the highest priority for selecting the functions.

P402, P403, and P404 can be used as IRQn-DS (n = 4, 14, 15) whether RTC inputs and tamper detection are selected or not. When using these interrupts, set the interrupt procedure after setting the VBTICTLR register. See [section 12.2.8. VBTICTLR : VBATT Input Control Register](#).

See [Figure 20.5](#).

The VBTICTLR register is only initialized by VBATT_POR reset. Therefore, when not using the RTC inputs and tamper detection, the associated bit of VBTICTLR register must be set to 0 after except VBATT_POR reset.

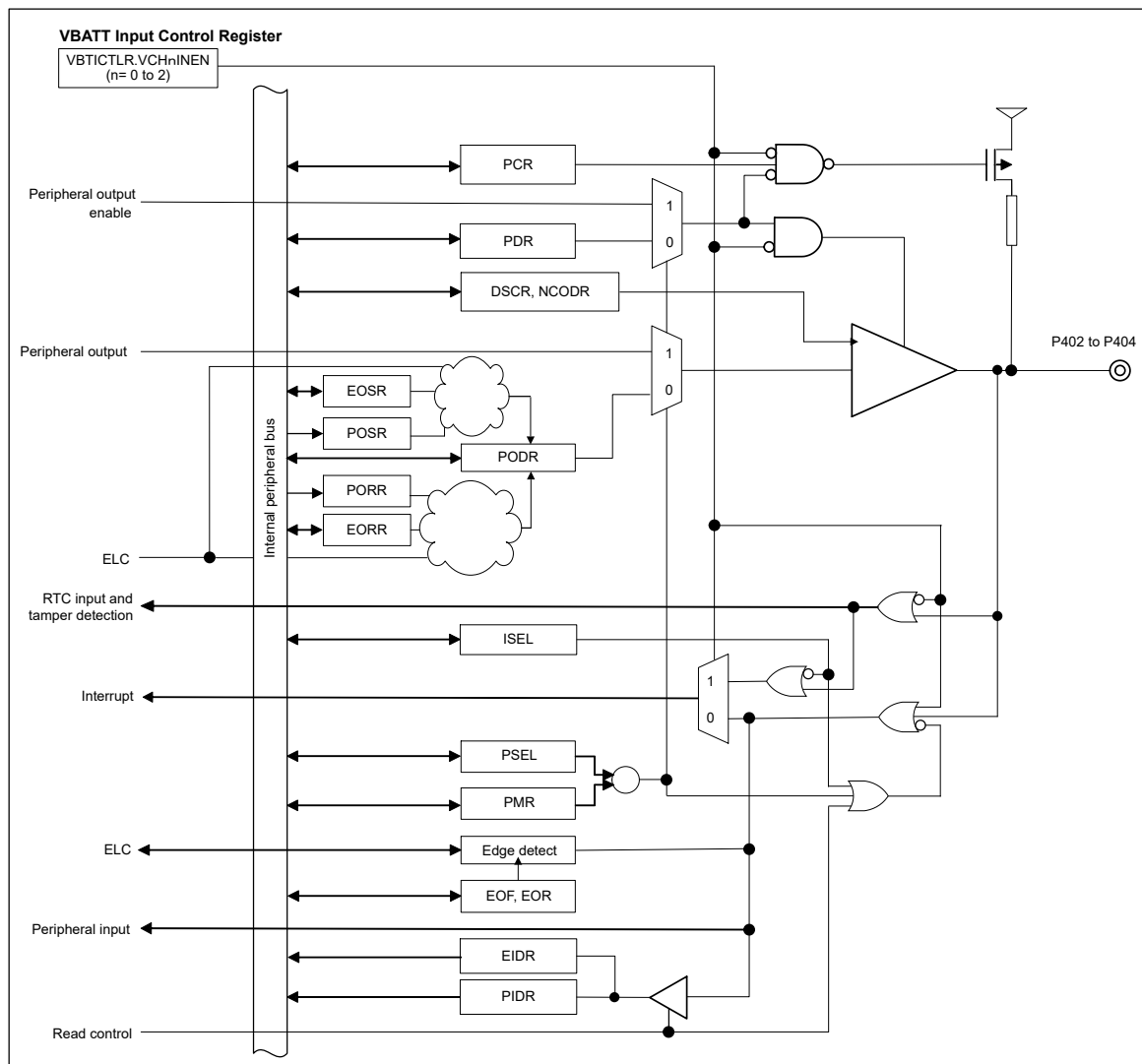


Figure 20.5 P402, P403, P404 diagram

20.5.6 Selecting Pins USB_DP and USB_DM

Pins USB_DP and USB_DM are shared with pins P814 and P815, respectively. Pins USB_DP and P814 can be set with the PFS.P814PFS.PMR bit, and pins USB_DM and P815 can be set with the PFS.P815PFS.PMR bit. Table 20.5 shows setting values of bits PFS.P814PFS.PMR and PFS.P815PFS.PMR with each selected pin.

Table 20.5 Selecting the USB/PORT pins

PMR Bits Settings		Pins Selected	
P814PFS.PMR bit	P815PFS.PMR bit	P814/USB_DP Pin	P815/USB_DM Pin
0	0	P814	P815
0	1	P814	P815
1	0	P814	P815
1	1	USB_DP	USB_DM

Note: When using P814/USB_DP and P815/USB_DM as GPIO pins (P814 and P815), use the USB related registers with their initial values.

Note: When using P814/USB_DP and P815/USB_DM as USB pins (USB_DP and USB_DM), use the GPIO related registers for P814 and P815 with their initial values.

Note: When using P814/USB_DP and P815/USB_DM as GPIO pins or USB pins, set these pins only once after a reset.

20.5.7 Pull-up/Pull-down Setting for P814 and P815 at USBFS/GPIO Function

When the GPIO function of P814 and P815 is used, the Pull-up/Pull-down function of USBFS register affects to P814 and P815. Therefore, before using GPIO function, the pull-up and pull-down control of USBFS needs to be disabled. The control register is SYSCFG.DPRPU and SYSCFG.DRPD bits.

20.5.8 Notes on using P215 and P214

Table 20.6 shows the pin selection and notes for P215 and P214.

Table 20.6 Pin selection and notes for P215 and P214

No.	P215	P214	Description
1	XCIN	XCOUT	Permitted*3
2	EXCIN*1	General purpose input*1	Permitted*4
3	EXCIN*1	Unused*2	Permitted*4
4	General purpose input*1	General purpose input*1	Permitted*5
5	General purpose input*1	Unused*2	Permitted*5
6	Unused*2	General purpose input*1	Permitted*6
7	Unused*2	Unused*2	Permitted*6

Note 1. Do not input a voltage higher than VCC to this pin when $VCC > V_{DET BATT}$.

Do not input a voltage higher than V_{BATT} to this pin when $VCC \leq V_{DET BATT}$.

Note 2. Unused pin should be connected to VSS through a resistor.

Note 3. Set the SOMCR.SOSEL bit to 0 and set SOSCCR.SOSTP bit to 0

Note 4. Set the SOMCR.SOSEL bit to 1 and set SOSCCR.SOSTP bit to 0

Note 5. Set the SOMCR.SOSEL bit to 0 and set SOSCCR.SOSTP bit to 1

Note 6. Set the SOSCCR.SOSTP bit to 1.

20.6 Peripheral Select Settings for Each Product

This section describes the pin function select configuration using the PmnPFS register. Some pin names have added _A, _B, or _C suffixes. When assigning IIC, SPI, and SDHI functionality, select the functional pins having the same suffix. The other pins can be selected regardless of the suffix. Assigning the same function to two or more pins simultaneously is prohibited.

1. In Pmn pin function select register(PmnPFS), the PSEL bits have to be set when the PMR bit of the target pin is 0. If the PSEL bits are set when the PMR bit is 1, the unexpected edges may be input at the input function or the unexpected pulses may be output to the external pin at the output function.
2. Only the allowed values (functions) should be specified in the PSEL bits of PmnPFS register. If a value which is not allowed for the register is specified, the correct operation is not guaranteed.
3. The single function should not be assigned to the multiple pins by PmnPFS register. When the GPT1, GPT5, SCI3, IIC0 or SPI0 are configured as secure and these pin function is being assigned to the pin which security attribution is set as secure by the PmSAR register, the write access to the PSEL bits for setting same function as secure pin in other pins is ignored when the security attribution of that pin is non-secure. For example, if the PSARE.PSARE30 bit is 0 (GPT1 is secure) and the P209PFS.PSEL bits is 00011b (pin function is GTIOC1A) and the P2SAR.P209SA bit is 0 (P209 is secure), the write 00011b to the P405PFS.PSEL bits is ignored when the P4SAR.P405SA bit is 1 (P405 is non-secure).
4. The PORT0, PORT5 and PORT8 have the analog functions such as A/D converter. When these pins are used as an analog function, for avoiding the loss of resolution, the PMR bit should be set to 0 and PDR bit should be set to 0. After that, ASEL bit should be set to 1

Table 20.7 Register settings for input/output pin function (PORT0)

PSEL [4:0]Settings	Function	Pin															
		P000	P001	P002	P003	P004	P005	P006	P007	P008	P009	P010	P011	P012	P013	P014	P015
00000b (value after reset)	Hi-z/ JTAG/SWD	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z
ASEL bit		AN000/ IVCMP 2	AN001/ IVCMP 3	AN002/ IVCMP 2	AN003/ IVCMP 3	AN004/ IVCMP 2	AN005/ IVCMP 3	AN006/ IVCMP 2	AN007/ IVCMP 3	AN008/ IVREF0	AN009/ IVREF1	AN010	AN011	AN012	AN013	AN014/ DA0/ IVCMP 0	AN015/ DA1/ IVCMP 0
ISEL bit		IRQ6- DS	IRQ7- DS	IRQ8- DS	IRQ29	IRQ9- DS	IRQ10- DS	IRQ11- DS	IRQ28	IRQ12- DS	IRQ13- DS	IRQ14	IRQ16	IRQ15	IRQ14	IRQ27	IRQ13
PCR bit	Pull-up	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
NCODR bit	N-ch Open-drain	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
DSCR[1:0] bit	Drive capability control*1	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L
289, 303 pins product		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
224 pins product		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	—	—	—	—	✓	✓
176 pins product		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	—	—	—	—	✓	✓

✓: Available

—: Setting prohibited

Note 1. The drive strength of this port cannot be controlled by PmnPFS.DSCR[1:0] bits.

Table 20.8 Register settings for input/output pin function (PORT1) (1 of 2)

PSEL [4:0]Settings	Function	Pin															
		P100	P101	P102	P103	P104	P105	P106	P107	P108	P109	P110	P111	P112	P113	P114	P115
00000b (value after reset)	Hi-z/JTAG/SWD	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z
00001b	AGT	AGTIO0	AGTE0	AGTO0	—	—	—	AGTOB0	AGTOA0	—	—	—	—	—	—	—	—
00010b	GPT*2	GTETRGA	GTETRGB	GTOWLO	GTOWUP	GTETRGB	—	GTOWLO	GTOWUP	—	—	—	—	GTETRGA	GTETRGB	GTETRGC	GTETRGD
00011b	GPT*2	GTIOC8B	GTIOC8A	GTIOC2B	GTIOC2A	GTIOC1B	GTIOC1A	GTIOC8B	GTIOC8A	GTIOC10B	GTIOC10A	GTIOC9B	GTIOC9A	GTIOC3B	GTIOC2A	GTIOC2B	GTIOC5A
00100b	SCI*1	—	—	—	—	—	CTS_RTS8_B/SS8_B	CTS8_B	CTS4_A	—	—	—	—	TXD0_A/SDA0_A/MOSIO_A	RXD0_A/SCL0_A/MISO0_A	CTS_RTS0_A/SS0_A	CTS0_A
00101b	SCI*1	SCK9_A	RXD9_A/SCL9_A/MISO9_A	TXD9_A/SDA9_A/MOSI9_A	CTS_RTS9_A/SS9_A	CTS9_A	—	—	—	—	—	—	—	—	—	—	—
00110b	SPI*1	MISOB_A	MOSIB_A	RSPCKB_A	SSLB0_A	SSLB1_A	SSLB2_A	SSLB3_A	—	—	—	—	—	SSLA2_B	SSLA1_B	SSLA0_B	MOSIA_B
01000b	DSMIF	DSM0CLK2	DSM0CLK1	DSM0CLK0	DSM0DAT2	DSM0DAT1	DSM0DAT0	—	—	—	—	—	—	—	—	—	—
01010b	CAC/ADC16H	—	—	ADTRG0	AD1FLAG1	AD0FLAG1	ADSYNC	ADST1	ADST0	—	—	—	—	ADST0	ADST1	ADSYNC	AD0FLAG1
01011b	BUS	—	—	—	—	—	—	—	—	—	—	—	—	D3/DQ3	D4/DQ4	D5/DQ5	D6/DQ6
01101b	SCI	—	—	—	—	—	DE8	—	—	—	—	—	—	—	—	DE0	—
01110b	SCI	DE9	—	—	DE9	—	—	—	—	—	—	—	—	—	—	—	—
10000b	CANFD	—	—	CRX0	CTX0	—	—	—	—	—	—	—	—	—	—	—	—
10101b	SDHI*1	—	—	—	—	—	—	—	—	SD0DAT6_C	SD0DAT5_C	SD0DAT4_C	SD0DAT3_C	SD0DAT4_B	SD0DAT5_B	SD0DAT6_B	SD0DAT7_B
10110b	ESWM (GMII/MII)	GPTP_MATC_H1	GPTP_CAPTURE1	—	GPTP_PPS0	GPTP_MATC_H0	GPTP_CAPTURE0	ET1_LI_NKSTA	ET1_I_NT	—	—	—	—	—	—	—	—
11010b	ESC	CAT12CDATA	CAT12CCLK	—	CATLATCH1	CATLATCH0	CATSYNC1	CAT1_LINKSTA	—	—	—	—	—	—	—	—	—

Table 20.8 Register settings for input/output pin function (PORT1) (2 of 2)

PSEL [4:0]Settings	Function	Pin															
		P100	P101	P102	P103	P104	P105	P106	P107	P108	P109	P110	P111	P112	P113	P114	P115
11100b	OSPI	OM_0_ SIO0	OM_0_ SIO3	OM_0_ SIO4	OM_0_ SIO2	OM_0_ CS1	OM_0_ ECSIN T1	OM_0_ RESET	OM_0_ CS0	—	—	—	—	—	—	—	—
11110b	ULPT	—	—	—	—	—	ULPT0 1-DS	ULPTE E1-DS	—	—	—	—	—	ULPT0 B0-DS	ULPT0 A0-DS	—	—
ASEL bit		—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
ISEL bit		IRQ2	IRQ1	IRQ17	IRQ16	IRQ1	IRQ0	IRQ16	IRQ31	IRQ24	IRQ23	IRQ20	IRQ19	IRQ27	IRQ28	IRQ30-DS	IRQ31-DS
PCR bit	Pull-up	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
NCODR bit	N-ch Open-drain	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
DSCR [1:0] bit	Drive capability control	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH
289, 303 pins product		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
224 pins product		✓	✓	✓	✓	✓	✓	✓	✓	—	—	—	—	✓	✓	✓	✓
176 pins product		✓	✓	✓	✓	✓	✓	✓	✓	—	—	—	—	✓	✓	✓	✓

✓ : Available

— : Setting prohibited

Note 1. Recommend using pins that have a letter appended to their names, for instance “_A”, “_B”, or “_C” to indicate group membership. For the interface, the AC portion of the electrical characteristics is measured for each group.

Note 2. There are 2 types output buffer which are middle drive and high drive. Recommend using same drive buffer for output skew spec (t_{GTISK}).

Table 20.9 Register settings for input/output pin function (PORT2) (1 of 2)

PSEL [4:0]Settings	Function	Pin															
		P200 ⁴	P201	P202	P203	P204	P205	P206	P207	P208	P209	P210	P211	P212	P213	P214	P215
00000b (value after reset)	Hi-z/ JTAG/SW D	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	TDI	TDO	TMS/ SWDIO	TCK/ SWCLK	Hi-z	Hi-z	Hi-z	Hi-z
00001b	AGT	—	—	—	—	AGTIO 1	AGTO1	—	—	—	—	—	—	AGTEE 1	—	—	—
00010b	GPT ²	—	—	—	—	GTIW	GTIV	GTIU	—	GTOVL O	GTOVU P	GTOUL O	GTOUJ P	GTETR GD	GTETR GC	—	—
00011b	GPT ²	—	—	GTIOC 5B	GTIOC 5A	GTIOC 4B	GTIOC 4A	GTCPPO 0	GTCPPO 3	GTIOC 1B	GTIOC 1A	GTIOC 0B	GTIOC 0A	GTIOC 0B	GTIOC 0A	—	—
00100b	SCI ¹	—	—	CTS_R TS4_A/ SS4_A	RXD4_A/ SCL4_A/ MISO4_A	SCK4_A	TXD4_A/ SDA4_A/ MOSI4_A	—	—	—	—	—	—	—	—	—	—
00101b	SCI ¹	—	—	—	—	—	—	—	—	RXD9_B/ SCL9_B/ MISO9_B	TXD9_B/ SDA9_B/ MOSI9_B	CTS_R TS9_B/ SS9_B	SCK9_B	RXD1_C/ SCL1_C/ MISO1_C	TXD1_C/ SDA1_C/ MOSI1_C	—	—
00110b	SPI ¹	—	—	MOSIA_A	RSPCK A_A	SSLA0_A	SSLA1_A	—	—	—	—	—	—	—	—	—	—
00111b	I3C/IIC ¹	—	—	—	—	SDA1_B	SCL1_B	—	—	—	—	—	—	—	—	—	—
01001b	RTC/ CLKOUT/ ACMPHS/ ETHPHYC LK	—	—	—	—	—	CLKOUT	CLKOUT	—	VCOUT	CLKOUT	—	—	—	—	—	—
01010b	CAC/ ADC16H	—	—	—	—	CACRE F	—	—	—	—	—	—	—	—	ADTRG 1	—	—
01011b	BUS	—	—	—	—	—	—	CS7	—	—	—	—	—	—	—	—	—
01101b	SCI	—	—	DE4	—	DE4	—	—	—	—	—	—	—	—	—	—	—
01110b	SCI	—	—	—	—	—	—	—	—	—	—	DE9	DE9	—	—	—	—
10000b	CANFD	—	—	CRX0	CTX0	—	—	—	—	CRX1	CTX1	—	—	—	—	—	—
10011b	USBFS	—	—	USB_E XICEN	USB_V BUSEN	USB_O VRCUR B	USB_O VRCUR A	USB_V BUSEN	—	—	—	—	—	—	—	—	—

Table 20.9 Register settings for input/output pin function (PORT2) (2 of 2)

PSEL [4:0]Settings	Function	Pin															
		P200 ^{*4}	P201	P202	P203	P204	P205	P206	P207	P208	P209	P210	P211	P212	P213	P214	P215
10101b	SDHI ^{*1}	—	—	SD1CM D_A	SD1CL K_A	SD1WP	SD1CD	SD0DA T7_C	—	—	—	—	—	—	—	—	—
10110b	ESWM (GMII/MII)	—	—	—	—	—	—	ET1_R X_DV	ET1_R XD5	—	—	—	—	—	—	—	—
10111b	ESWM (RMII)	—	—	—	—	—	—	RMII1 CRS_D V	—	—	—	—	—	—	—	—	—
11000b	ESWM (RGMII)	—	—	—	—	—	—	RGMII1 RX_C TL	—	—	—	—	—	—	—	—	—
11010b	ESC	—	—	—	—	—	—	CAT1_ RX_DV	—	—	—	—	—	—	—	—	—
11110b	ULPT	—	—	ULPTO B1	ULPTO A1	—	—	ULPTO B1	—	—	—	—	—	—	ULPTE E0	—	—
Don't care	Trace (Debug)/ Clock	—	—	—	—	—	—	—	—	—	SWO ^{*5}	—	—	EXTAL	XTAL	XCOUT	XCIN/ EXCIN
ASEL bit		—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
ISEL bit		NMI	IRQ4	IRQ3- DS	IRQ2- DS	IRQ26	IRQ1- DS	IRQ0- DS	IRQ25	IRQ3	IRQ25	IRQ24	IRQ23	IRQ3	IRQ2	IRQ21	IRQ20
PCR bit	Pull-up	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
NCODR bit	N-ch Open-drain	—	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	—	—
DSCR[1:0] bit	Drive capability control	—	L ^{*3}	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	—	—
289, 303 pins product		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
224 pins product		✓	✓	✓	✓	✓	✓	✓	—	✓	✓	✓	✓	✓	✓	✓	✓
176 pins product		✓	✓	—	—	—	—	✓	—	✓	✓	✓	✓	✓	✓	✓	✓

✓: Available

—: Setting prohibited

Note 1. Recommend using pins that have a letter appended to their names, for instance “_A”, “_B”, or “_C” to indicate group membership. For the interface, the AC portion of the electrical characteristics is measured for each group.

Note 2. There are 2 types output buffer which are middle drive and high drive. Recommend using same drive buffer for output skew spec (t_{GTSK}).

Note 3. The driver strength of this port is controlled by PmnPFS.DSCR[1:0] bits. Set the PmnPFS.DSCR[1:0] bits to Low Drive (00b).

Note 4. When using NMI pin interrupt, Port related registers setting are not required.

Note 5. To use this pin function, select the CoreSight TPIU.SPPR.PProtocol bits to SWO and set the TRPORTCR.OE bit to 1. The drive strength of this pin function is controlled by TRPORTCR.DRV[1:0] bits.

Table 20.10 Register settings for input/output pin function (PORT3) (1 of 2)

PSEL [4:0]Settings	Function	Pin															
		P300	P301	P302	P303	P304	P305	P306	P307	P308	P309	P310	P311	P312	P313	P314	P315
00000b (value after reset)	Hi-z/ JTAG/SW D	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z
00001b	AGT	—	AGTIO 0	—	—	—	—	—	—	—	—	AGTEE 1	AGTOB 1	AGTOA 1	—	—	—
00010b	GPT ^{*2}	—	GTOUL O	GTOUU P	—	GTOVL O	GTOVU P	GTIW	GTIV	GTIU	—	—	GTADS M1	GTADS M0	—	—	—
00011b	GPT ^{*2}	GTIOC 3A	GTIOC 4B	GTIOC 4A	GTIOC 7B	GTIOC 7A	GTCPP O12	GTCPP O11	GTCPP O10	GTCPP O9	GTCPP O8	GTCPP O7	GTCPP O6	GTCPP O5	—	—	—
00100b	SCI ^{*1}	SCK0_ A	TXD6_ B/ SDA6_ B/ MOSI6_ B	RXD6_ B/ SCL6_ B/ MISO6_ B	SCK6_ B	—	—	—	CTS_R TS6_A/ SS6_A	—	—	—	—	—	—	—	—
00101b	SCI ^{*1}	—	—	—	—	—	—	—	—	CTS3_ B	CTS9_ B	TXD3_ B/ SDA3_ B/ MOSI3_ B	SCK3_ B	CTS_R TS3_B/ SS3_B	TXD3_ C/ SDA3_ C/ MOSI3_ C	RXD3_ C/ SCL3_ C/ MISO3_ C	SCK3_ C
00110b	SPI ^{*1}	SSLA3_ B	—	—	—	—	—	—	—	—	—	—	—	—	MISOA_ A	SSLA2_ A	SSLA3_ A

Table 20.10 Register settings for input/output pin function (PORT3) (2 of 2)

PSEL [4:0]Settings	Function	Pin															
		P300	P301	P302	P303	P304	P305	P306	P307	P308	P309	P310	P311	P312	P313	P314	P315
01001b	RTC/ CLKOUT/ ACMPHS/ ETHPHYC LK	—	—	—	—	—	—	—	—	ETHPH YCLK	VCOUT	—	—	—	—	—	—
01010b	CAC/ ADC16H	—	—	—	—	—	—	—	—	—	—	—	—	—	—	ADTRG 0	—
01011b	BUS	D2/DQ2	D1/DQ1	D0/DQ0	—	—	—	—	—	—	—	—	—	—	—	—	—
01101b	SCI	DE0	—	—	DE6	—	—	—	DE6	—	—	—	—	—	—	—	—
01110b	SCI	—	—	—	—	—	—	—	—	—	—	—	DE3	DE3	—	—	DE3
10000b	CANFD	—	—	—	—	—	—	—	—	—	—	—	CRX0	CTX0	—	—	—
10011b	USBFS	—	—	—	—	—	—	—	—	—	—	—	—	—	USB_ID	—	—
10101b	SDHI ¹	SD0DA T3_B	SD0DA T2_B	SD0DA T1_B	—	SD0DA T0_B	SD0WP	SD0CD	SD0CM D_B	SD0CL K_B	—	—	—	—	SD1DA T0_A	SD1DA T1_A	—
10110b	ESWM (GMII/MII)	—	—	—	—	ET1_T XD3	ET1_T XD2	ET1_T XD1	ET1_T XD0	ET1_T X_ER	ET1_G TX_CL K	ET1_T X_EN	ET1_T X_CLK	ET1_R X_ER	—	—	—
10111b	ESWM (RMII)	—	—	—	—	—	—	RMII1_ TXD1	RMII1_ TXD0	—	—	RMII1_ TX_EN	—	RMII1_ RX_ER	—	—	—
11000b	ESWM (RGMII)	—	—	—	—	RGMII1_ TXD3	RGMII1_ TXD2	RGMII1_ TXD1	RGMII1_ TXD0	—	RGMII1_ TXC	RGMII1_ TX_C TL	—	—	—	—	—
11010b	ESC	—	—	—	—	CAT1_ ETXD3	CAT1_ ETXD2	CAT1_ ETXD1	CAT1_ ETXD0	—	—	CAT1_ TX_EN	CAT1_ TX_CL K	CAT1_ RX_ER	—	—	—
11110b	ULPT	ULPTE VIO-DS	ULPTE E0-DS	ULPTO 0-DS	—	ULPTO 1	ULPTE E1	ULPTE V1	ULPTO A1	ULPTO B1	—	—	—	—	—	—	—
Don't care	Trace (Debug)	—	—	—	—	TDATA 3 ³	TDATA 2 ³	TDATA 1 ³	TDATA 0 ³	TCLK ³	—	—	—	—	—	—	—
ASEL bit		—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
ISEL bit		IRQ4	IRQ6	IRQ5	IRQ29- DS	IRQ9	IRQ8	IRQ28- DS	IRQ27- DS	IRQ26- DS	IRQ25- DS	IRQ24- DS	IRQ23- DS	IRQ22- DS	IRQ27	IRQ28	IRQ29
PCR bit	Pull-up	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
NCODR bit	N-ch Open-drain	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
DSCR[1:0] bit	Drive capability control	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH
289, 303 pins product		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
224 pins product		✓	✓	✓	—	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
176 pins product		✓	✓	✓	—	✓	✓	✓	✓	✓	—	✓	✓	✓	—	—	—

✓: Available

—: Setting prohibited

Note 1. Recommend using pins that have a letter appended to their names, for instance “_A”, “_B”, or “_C” to indicate group membership. For the interface, the AC portion of the electrical characteristics is measured for each group.

Note 2. There are 2 types output buffer which are middle drive and high drive. Recommend using same drive buffer for output skew spec (t_{GTISK}).

Note 3. To use this pin function, select the CoreSight TPIU.SPPR.PProtocol bits to parallel port and set the TRPORTCR.OE bit to 1. The drive strength of this pin function is controlled by TRPORTCR.DRV[1:0] bits. How many pins of Trace Data are used can be controlled by TRPORTSZ.PORTSIZE[31:0] bits.

Table 20.11 Register settings for input/output pin function (PORT4) (1 of 2)

PSEL [4:0]Settings	Function	Pin															
		P400	P401	P402	P403	P404	P405	P406	P407	P408	P409	P410	P411	P412	P413	P414	P415
00000b (value after reset)	Hi-z/ JTAG/SW D	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z
00001b	AGT	AGTIO 1	—	—	—	—	AGTIO 1	—	AGTIO 0	—	—	AGTOB 1	AGTOA 1	AGTEE 1	—	—	—
00010b	GPT ²	—	GTETRA GA	—	—	—	—	—	—	GTOW LO	GTOW UP	GTOVL O	GTOVU P	GTOUL O	GTOUU P	—	—
00011b	GPT ²	GTIOC 6A	GTIOC 6B	—	GTIOC 3A	GTIOC 3B	GTIOC 1A	GTIOC 1B	GTIOC 10B	GTIOC 10A	—	GTIOC 9B	GTIOC 9A	GTCPP 08	GTCPP 07	GTIOC 0B	GTIOC 0A

Table 20.11 Register settings for input/output pin function (PORT4) (2 of 2)

PSEL [4:0]Settings	Function	Pin															
		P400	P401	P402	P403	P404	P405	P406	P407	P408	P409	P410	P411	P412	P413	P414	P415
00100b	SCI ^{*1}	—	—	—	—	—	SCK2_ B	TXD2_ B/ SDA2_ B/ MOSI2_ B	—	—	—	—	—	—	—	RXD4_ B/ SCL4_ B/ MISO4_ B	TXD4_ B/ SDA4_ B/ MOSI4_ B
00101b	SCI ^{*1}	TXD1_ A/ SDA1_ A/ MOSI1_ A	RXD1_ A/ SCL1_ A/ MISO1_ A	SCK1_ A	CTS_ R TS1_ A/ SS1_ A	CTS1_ A	—	—	SCK1_ C	RXD3_ A/ SCL3_ A/ MISO3_ A	TXD3_ A/ SDA3_ A/ MOSI3_ A	SCK3_ A	CTS_ R TS3_ A/ SS3_ A	CTS3_ A	—	—	—
00110b	SPI ^{*1}	—	—	—	—	—	—	SSLA3_ C	—	—	—	—	—	—	—	SSLB0_ B	RSPCK_ B
00111b	I3C/IIC ^{*1}	I3C_ SC L0	I3C_ SD A0	—	—	—	—	—	SDA0_ B	SCL0_ B	SDA0_ A	SCL0_ A	—	—	—	—	—
01000b	DSMIF	—	—	—	—	DSM0D AT0	DSM0D AT1	DSM0D AT2	—	—	—	—	—	—	—	—	—
01001b	RTC/ CLKOUT/ ACMPHS/ ETHPHYC LK	—	—	—	—	—	—	—	RTCCOU T	—	—	—	—	—	—	—	—
01010b	CAC/ ADC16H	ADTRG 1	—	CACRE F	AD0FL AG1	AD1FL AG1	—	—	ADTRG 0	ADSYN C	ADST1	ADST0	CACRE F	—	—	—	—
01011b	BUS	—	—	—	—	—	—	—	CS6	A17	A18	A19	A20	A21	A22	A23	WAIT
01101b	SCI	—	—	—	—	—	DE2	—	—	—	—	—	—	—	—	—	—
01110b	SCI	—	—	DE1	DE1	—	—	—	DE1	—	—	DE3	DE3	—	—	—	—
10000b	CANFD	—	CTX0	CRX0	—	—	—	—	—	—	—	—	—	—	—	CRX1	CTX1
10011b	USBFS	—	—	—	—	—	—	—	USB_ V BUS	USB_ V BUSEN	USB_ O VRCUR A-DS	USB_ O VRCUR B-DS	USB_ ID	USB_ E XICEN	—	—	—
10101b	SDHI ^{*1}	SD1CL K_ B	SD1CM D_ B	SD1DA T0_ B	SD1DA T1_ B	SD1DA T2_ B	SD1DA T3_ B	SD1CD	—	—	—	—	—	—	—	—	—
10110b	ESWM (GMII/MII)	—	—	ET0_ LI NKSTA	ET1_ W OL	ET0_ W OL	ET0_ R X_ DV	ET0_ R XD3	GPTP_ PTPOU T3	GPTP_ PTPOU T2	GPTP_ CAPTU RE0	GPTP_ MATCH 0	GPTP_ PTPOU T1	GPTP_ PTPOU T0	ET_ TA S_ STA 3	ET1_ M DIO	ET1_ M DC
10111b	ESWM (RMII)	—	—	—	—	—	RMII0_ CRS_ D V	—	—	—	—	—	—	—	—	—	—
11000b	ESWM (RGMII)	—	—	—	—	—	RGMII0_ RX_ C TL	RGMII0_ RXD3	—	—	—	—	—	—	—	—	—
11010b	ESC	—	—	CAT0_ L INKSTA	CAT12C CLK	CAT12C DATA	CAT0_ RX_ DV	CAT0_ ERXD3	CATSY NC1	CATRE SETOU T	CAT12C DATA	CAT12C CLK	—	—	—	CATLA TCH1	CATLA TCH0
11110b	ULPT	—	—	—	—	—	—	—	—	ULPT0 B0	ULPT0 A0	—	—	—	—	ULPTE E1	—
Don't care	RTC	—	—	RTCIC0 *3	RTCIC1 *3	RTCIC2 *3	—	—	—	—	—	—	—	—	—	—	—
ASEL bit		—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
ISEL bit		IRQ0	IRQ5- DS	IRQ4- DS	IRQ14- DS	IRQ15- DS	IRQ30	IRQ31	IRQ22	IRQ7	IRQ6	IRQ5	IRQ4	IRQ20- DS	IRQ18	IRQ9	IRQ8
PCR bit	Pull-up	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
NCODR bit	N-ch Open-drain	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
DSCR[1:0] bit	Drive capability control	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH
289, 303 pins product		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
224 pins product		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	—	—	✓	✓
176 pins product		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	—	—	—	✓	✓

✓: Available

—: Setting prohibited

Note 1. Recommend using pins that have a letter appended to their names, for instance “_A”, “_B”, or “_C” to indicate group membership. For the interface, the AC portion of the electrical characteristics is measured for each group.

Note 2. There are 2 types output buffer which are middle drive and high drive. Recommend using same drive buffer for output skew spec (t_{GTISK}).

Note 3. To use this pin function, set the associated pin as a general input (set the PmnPFS.PDR and PmnPFS.PMR bits to 0).

Table 20.12 Register settings for input/output pin function (PORT5)

PSEL [4:0]Settings	Function	Pin															
		P500	P501	P502	P503	P504	P505	P506	P507	P508	P509	P510	P511	P512	P513	P514	P515
00000b (value after reset)	Hi-z/ JTAG/SWD	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z
00010b	GPT ²	—	—	—	GTOUUP	GTOULO	GTOWUP	GTOWLO	GTADSM0	GTADSM1	—	—	—	—	—	—	—
00011b	GPT ²	GTIOC11B	GTIOC12A	GTIOC12B	GTCPP06	GTCPP01	GTCPP02	GTCPP03	GTIOC0A	GTIOC0B	GTIOC1A	GTIOC1B	GTIOC0B	GTIOC0A	GTIOC13B	GTIOC13B	GTIOC13A
00100b	SCI ¹	RXD8_B/ SCL8_B/ MISO8_B	TXD8_B/ SDA8_B/ MOSI8_B	SCK8_B	—	—	—	—	—	—	—	—	CTS_RTS8_A/ SS8_A	CTS8_A	SCK8_A	SCK4_C	CTS_RTS4_C/ SS4_C
00101b	SCI ¹	—	—	—	—	—	—	—	CTS_RTS7_A/ SS7_A	CTS5_A	CTS_RTS5_A/ SS5_A	RXD5_A/ SCL5_A/ MISO5_A	—	—	—	—	—
00111b	I3C/IIC ¹	—	—	—	—	—	—	—	—	—	—	—	SDA1_A	SCL1_A	—	SDA2_B	SCL2_B
01010b	CAC/ ADC16H	CACRE F	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
01011b	BUS	—	—	—	A6	A7	A8	A9	A10	A11	A12	A13	—	—	—	—	—
01101b	SCI	—	—	DE8	—	—	—	—	—	—	—	—	DE8	—	DE8	DE4	DE4
01110b	SCI	—	—	—	—	—	—	—	DE7	—	DE5	—	—	—	—	—	—
10000b	CANFD	—	—	—	—	—	—	—	—	—	—	—	CRX1	CTX1	—	—	—
10011b	USBFS	USB_V BUSEN	USB_O VRCUR A	USB_O VRCUR B	—	—	—	—	—	—	—	—	—	—	—	—	—
10101b	SDHI ¹	SD1DAT5_A	SD1DAT6_A	SD1DAT7_A	SD0CD	SD0WP	SD0CLK_A	SD0CMD_A	—	—	—	—	—	—	—	—	—
10110b	ESWM (GMII/MII)	—	—	—	—	—	—	—	ET_TAS_STA0	ET_TAS_STA1	ET_TAS_STA2	ET_TAS_STA3	ET1_LI NKSTA	ET1_IN T	ET0_IN T	ET_TAS_STA1	ET_TAS_STA0
11010b	ESC	—	—	—	—	—	—	—	—	—	—	—	CAT1_L INKSTA	—	—	—	—
11110b	ULPT	—	—	—	—	—	—	—	—	—	ULPTE VI1	ULPTE VI0	—	—	—	—	—
ASEL bit		AN021	AN020	AN019	—	—	—	—	—	—	—	—	—	—	AN016/ IVCMP0	—	—
ISEL bit		IRQ24	IRQ25	IRQ26	IRQ6	IRQ7	IRQ8	IRQ9	IRQ10	IRQ1	IRQ2	IRQ3	IRQ15	IRQ14	IRQ31	IRQ13	IRQ12
PCR bit	Pull-up	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
NCODR bit	N-ch Open-drain	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
DSCR[1:0] bit	Drive capability control	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH
289, 303 pins product		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
224 pins product		—	—	—	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	—	—	—
176 pins product		—	—	—	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	—	—	—

✓: Available

—: Setting prohibited

Note 1. Recommend using pins that have a letter appended to their names, for instance “_A”, “_B”, or “_C” to indicate group membership. For the interface, the AC portion of the electrical characteristics is measured for each group.

Note 2. There are 2 types output buffer which are middle drive and high drive. Recommend using same drive buffer for output skew spec (t_{GTISK}).

Table 20.13 Register settings for input/output pin function (PORT6)

PSEL [4:0]Settings	Function	Pin															
		P600	P601	P602	P603	P604	P605	P606	P607	P608	P609	P610	P611	P612	P613	P614	P615
00000b (value after reset)	Hi-z/ JTAG/SW D	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z
00001b	AGT	—	—	—	—	—	—	—	—	—	—	—	—	—	AGTO1	AGTO0	—
00010b	GPT ²	—	—	—	—	—	—	—	—	GTOW UP	GTIU	GTOU P	GTOUL O	—	GTETR GA	GTETR GB	GTETR GC
00011b	GPT ²	GTIOC 6B	GTIOC 6A	GTIOC 7B	GTIOC 7A	GTIOC 8B	GTIOC 8A	—	—	GTCP O4	GTIOC 5B	GTIOC 4A	GTIOC 4B	GTIOC 9A	GTIOC 9B	GTCP O9	GTCP O10
00100b	SCI ¹	—	SCK0_ B	RXD0_ B/ SCL0_ B/ MISO0_ B	TXD0_ B/ SDA0_ B/ MOSI0_ B	CTS_R TS0_ B/ SS0_ B	CTS0_ B	—	—	—	TXD0_ C/ SDA0_ C/ MOSI0_ C	RXD0_ C/ SCL0_ C/ MISO0_ C	SCK0_ C	CTS_R TS0_ C/ SS0_ C	CTS0_ C	—	—
00101b	SCI ¹	—	—	—	—	—	—	—	—	TXD5_ A/ SDA5_ A/ MOSI5_ A	—	—	—	—	—	RXD7_ A/ SCL7_ A/ MISO7_ A	TXD7_ A/ SDA7_ A/ MOSI7_ A
00110b	SPI ¹	—	—	—	—	—	—	—	—	—	MISOA_ B	RSPCK A_B	MOSIA_ B	SSLA0_ B	—	—	—
01001b	RTC/ CLKOUT/ ACMPHS/ ETHPHYC LK	—	RTCOU T	—	—	—	—	—	—	—	—	—	CLKOU T	—	—	—	—
01010b	CAC/ ADC16H	CACRE F	—	—	—	—	—	—	—	CACRE F	AD1FL AG1	—	CACRE F	—	—	—	—
01011b	BUS	—	—	—	—	—	—	WR3/B C3	D31/ DQ31	A14	D7/DQ7	D12/ DQ12	D13/ DQ13	D14/ DQ14	D15/ DQ15	WR/WR 0/ DQM0	WR2/B C2/ DQM2
01101b	SCI	—	DE0	—	—	DE0	—	—	—	—	—	—	DE0	DE0	—	—	—
10000b	CANFD	—	—	—	—	—	—	—	—	—	CTX1	CRX1	—	—	—	—	—
10110b	ESWM (GMII/MII)	ET1_W OL	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
11100b	OSPI	OM_0_ RSTO1	OM_0_ WP1	OM_1_ SCLKN	OM_1_ SCLK	OM_1_ SIO7	OM_1_ SIO1	OM_1_ SIO6	OM_1_ DQS	—	—	—	—	—	—	—	—
11110b	ULPT	ULPTE VI1-DS	ULPTE VI0	ULPTE E0	ULPTO 0	—	—	—	—	—	ULPTO A1-DS	ULPTO B1-DS	—	—	—	—	—
ASEL bit		—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
ISEL bit		IRQ30	IRQ29	IRQ28	IRQ27	IRQ26	IRQ25	IRQ24	IRQ23	IRQ22	IRQ29	IRQ16	IRQ17	IRQ18	IRQ19	IRQ20	IRQ7
PCR bit	Pull-up	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
NCODR bit	N-ch Open-drain	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
DSCR[1:0] bit	Drive capability control	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH
289, 303 pins product ³		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
224 pins product		✓	✓	✓	✓	✓	—	—	—	✓	✓	✓	✓	✓	✓	✓	✓
176 pins product		✓	✓	—	—	—	—	—	—	✓	✓	✓	✓	✓	✓	✓	✓

✓: Available

—: Setting prohibited

Note 1. Recommend using pins that have a letter appended to their names, for instance “_A”, “_B”, or “_C” to indicate group membership. For the interface, the AC portion of the electrical characteristics is measured for each group.

Note 2. There are 2 types output buffer which are middle drive and high drive. Recommend using same drive buffer for output skew spec (t_{GTISK}).

Note 3. For BGA303 package, P603 to P607 are connected to the SiP flash inside the package. To access SiP Flash memory, set OSPI in PSEL settings.

Table 20.14 Register settings for input/output pin function (PORT7) (1 of 2)

PSEL [4:0]Settings	Function	Pin															
		P700	P701	P702	P703	P704	P705	P706	P707	P708	P709	P710	P711	P712	P713	P714	P715
00000b (value after reset)	Hi-z/ JTAG/SW D	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z

Table 20.14 Register settings for input/output pin function (PORT7) (2 of 2)

PSEL [4:0]Settings	Function	Pin															
		P700	P701	P702	P703	P704	P705	P706	P707	P708	P709	P710	P711	P712	P713	P714	P715
00001b	AGT	—	—	—	AGTO1	AGTO0	AGTIO0	AGTIO0	—	—	—	—	AGTEE0	AGTOB0	AGTOA0	—	—
00010b	GPT ²	—	—	—	—	GTADS M0	GTADS M1	—	—	—	—	—	—	—	—	—	—
00011b	GPT ²	GTIOC 5A	GTIOC 5B	GTIOC 6A	GTIOC 6B	GTCPP 00	GTCPP 01	GTCPP 02	GTCPP 03	GTCPP 06	GTCPP 05	GTIOC 11B	GTIOC 11A	GTIOC 2B	GTIOC 2A	GTIOC 12B	GTIOC 12A
00100b	SCI ¹	RXD2_ B/ SCL2_ B/ MISO2_ B	CTS_R TS2_ B/ SS2_ B	CTS2_ B	—	—	—	—	—	SCK4_ B	CTS_R TS4_ B/ SS4_ B	CTS4_ B	—	—	CTS4_ C	TXD4_ C/ SDA4_ C/ SCL4_ C/ MISO4_ C	RXD4_ C/ SCL4_ C/ MISO4_ C
00101b	SCI ¹	—	—	—	—	—	CTS1_ B	RXD1_ B/ SCL1_ B/ MISO1_ B	TXD1_ B/ SDA1_ B/ MOSI1_ B	—	—	—	CTS_R TS1_ C/ SS1_ C	CTS1_ C	—	—	—
00110b	SPI ¹	MISOA_ C	MOSIA_ C	RSPCK A_ C	SSLA0_ C	SSLA1_ C	SSLA2_ C	—	—	MOSIB_ B	MISOB_ B	SSLB3_ B	SSLB2_ B	SSLB1_ B	—	—	—
00111b	I3C/IIC ¹	—	—	—	—	—	—	—	—	SDA2_ A	SCL2_ A	—	—	—	—	—	—
01000b	DSMIF	DSM0C LK0	DSM0C LK1	DSM0C LK2	DSM1D AT0	DSM1D AT1	DSM1D AT2	DSM1C LK0	DSM1C LK1	—	—	—	—	—	—	—	—
01001b	RTC/ CLKOUT/ ACMPHS/ ETHPHYC LK	—	—	—	VCOUT	—	—	ETHPHYCLK	ETHPHYCLK	—	—	—	—	—	—	—	—
01010b	CAC/ ADC16H	—	—	—	—	—	—	—	—	CACRE F	—	—	—	—	—	—	—
01011b	BUS	—	—	—	—	—	—	—	—	WR1/B C1	CS4	CS5	—	—	—	—	—
01101b	SCI	—	DE2	—	—	—	—	—	—	DE4	DE4	—	—	—	—	—	—
01110b	SCI	—	—	—	—	—	—	—	—	—	—	—	DE1	—	—	—	—
10000b	CANFD	—	—	—	—	CTX0	CRX0	—	—	—	—	—	—	—	—	—	—
10101b	SDHI ¹	SD1WP	SD1DA T4_ B	SD1DA T5_ B	SD1DA T6_ B	SD1DA T7_ B	—	—	—	—	—	—	—	—	—	—	—
10110b	ESWM (GMII/MII)	ET0_R XD2	ET0_R XD1	ET0_R XD0	ET0_R X_CLK	ET0_R X_ER	ET0_T X_EN	ET0_G TX_CLK	ET0_T X_ER	ET0_M DC	ET0_M DIO	ET0_LI NKSTA	GPTP_PPS0	GPTP CAPTURE1	GPTP MATCH 1	GPTP PPS1	ET_TA S_STA 2
10111b	ESWM (RMII)	—	RMII0_RXD1	RMII0_RXD0	RMII0_REF50 CK	RMII0_RX_ER	RMII0_TX_EN	—	—	—	—	—	—	—	—	—	—
11000b	ESWM (RGMII)	RGMII0_RXD2	RGMII0_RXD1	RGMII0_RXD0	RGMII0_RXC	—	RGMII0_TX_C TL	RGMII0_TXC	—	—	—	—	—	—	—	—	—
11010b	ESC	CAT0_ERXD2	CAT0_ERXD1	CAT0_ERXD0	CAT0_RX_CLK	CAT0_RX_ER	CAT0_TX_EN	—	—	CAT0_MDC	CAT0_MDIO	CAT0_L INKSTA	CATRE SETOUT	CATLA TCH1	CATLA TCH0	CATSY NC1	—
11110b	ULPT	—	ULPT0 1	ULPT0 0	—	—	—	—	—	—	—	—	—	—	—	—	—
ASEL bit		—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
ISEL bit		IRQ16-DS	IRQ17-DS	IRQ18-DS	IRQ19-DS	IRQ26	IRQ19	IRQ7	IRQ8	IRQ11	IRQ10	IRQ17	IRQ3	IRQ2	IRQ14	IRQ13	IRQ12
PCR bit	Pull-up	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
NCODR bit	N-ch Open-drain	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
DSCR[1:0] bit	Drive capability control	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH
289, 303 pins product		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
224 pins product		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	—
176 pins product		✓	✓	✓	✓	✓	✓	—	✓	✓	✓	—	—	—	—	—	—

✓ : Available

— : Setting prohibited

Note 1. Recommend using pins that have a letter appended to their names, for instance “_A”, “_B”, or “_C” to indicate group membership. For the interface, the AC portion of the electrical characteristics is measured for each group.

Note 2. There are 2 types output buffer which are middle drive and high drive. Recommend using same drive buffer for output skew spec (t_{GTISK}).

Table 20.15 Register settings for input/output pin function (PORT8)

PSEL [4:0]Settings	Function	Pin															
		P800	P801	P802	P803	P804	P805	P806	P807	P808	P809	P810	P811	P812	P813	P814	P815
00000b (value after reset)	Hi-z/ JTAG/SWD	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z
00001b	AGT	AGTOA0	AGTOB0	—	—	—	—	—	—	—	—	—	—	—	—	—	—
00010b	GPT ²	GTIU	GTIV	GTIW	GTETRG	GTETRGD	—	—	—	—	—	—	—	—	GTETRGA	—	—
00011b	GPT ²	GTIOC11A	GTIOC11B	GTIOC12A	GTIOC12B	GTIOC13A	—	—	GTIOC13A	GTIOC13B	—	GTIOC10A	GTIOC10B	GTIOC11A	GTIOC7B	GTIOC8B	GTIOC8A
00100b	SCI ¹	CTS2_A	TXD2_A/ SDA2_A/ MISO2_A	RXD2_A/ SCL2_A/ MISO2_A	SCK2_A	CTS_R TS2_A/ SS2_A	TXD8_A/ SDA8_A/ MISO8_A	RXD8_A/ SCL8_A/ MISO8_A	—	—	—	—	—	—	—	—	—
00101b	SCI ¹	—	—	—	—	—	—	—	—	RXD7_B/ SCL7_B/ MISO7_B	TXD7_B/ SDA7_B/ MISO7_B	SCK7_B	CTS7_B	CTS_R TS7_B/ SS7_B	SCK7_A	—	—
01000b	DSMIF	DSM1D AT0	DSM1D AT1	DSM1D AT2	DSM1C LK0	DSM1C LK1	—	—	—	DSM1C LK2	—	—	—	—	—	—	—
01011b	BUS	—	—	—	—	—	—	—	—	—	—	—	—	—	SDCS	—	—
01101b	SCI	—	—	—	DE2	DE2	—	—	—	—	—	—	—	—	—	—	—
01110b	SCI	—	—	—	—	—	—	—	—	—	—	DE7	—	DE7	DE7	—	—
10000b	CANFD	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CRX0	CTX0
10011b	USBFS	—	—	—	—	—	—	—	—	—	—	—	USB_ID	USB_E XICEN	—	USB_D P	USB_D M
10101b	SDHI ¹	—	—	—	—	—	—	—	—	—	—	SD1DA T2_A	SD1DA T3_A	SD1DA T4_A	—	—	—
10110b	ESWM (GMII/MII)	—	GPTP PPS1	—	—	—	ET1_M DIO	ET1_M DC	—	—	—	—	—	—	—	—	—
11010b	ESC	—	CATRE SETOUT	—	—	—	—	—	—	—	—	—	—	—	—	—	—
11100b	OSPI	OM_0 SIO5	OM_0 DQS	OM_0 SIO6	OM_0 SIO1	OM_0 SIO7	—	—	—	OM_0 SCLK	OM_0 SCLKN	—	—	—	—	—	—
11110b	ULPT	—	—	—	—	—	—	—	—	—	—	ULPTO A0	ULPTO B0	—	—	—	—
Don't care	USBFS	—	—	—	—	—	—	—	—	—	—	—	—	—	—	USB_D P	USB_D M
ASEL bit		—	—	—	—	—	AN017/ IVCMP0	AN018	—	—	—	—	—	AN022	—	—	—
ISEL bit		IRQ11	IRQ12	IRQ18	IRQ19	IRQ14	IRQ30	IRQ0	IRQ11	IRQ15	IRQ20	IRQ21	IRQ22	IRQ23	IRQ15	IRQ16	IRQ15
PCR bit	Pull-up	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
NCODR bit	N-ch Open-drain	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
DSCR[1:0] bit	Drive capability control	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH
289, 303 pins product		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
224 pins product		✓	✓	✓	✓	✓	—	—	—	✓	✓	—	—	—	✓	✓	✓
176 pins product		✓	✓	✓	✓	✓	—	—	—	✓	✓	—	—	—	✓	✓	✓

✓: Available

—: Setting prohibited

Note 1. Recommend using pins that have a letter appended to their names, for instance “_A”, “_B”, or “_C” to indicate group membership. For the interface, the AC portion of the electrical characteristics is measured for each group.

Note 2. There are 2 types output buffer which are middle drive and high drive. Recommend using same drive buffer for output skew spec (t_{GTISK}).

Table 20.16 Register settings for input/output pin function (PORT9)

PSEL [4:0]Settings	Function	Pin															
		P900	P901	P902	P903	P904	P905	P906	P907	P908	P909	P910	P911	P912	P913	P914	P915
00000b (value after reset)	Hi-z/ JTAG/SW D	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z
00001b	AGT	—	AGTIO 1	—	—	—	—	—	—	—	—	—	—	—	—	—	—
00010b	GPT ²	GTADS M0	GTADS M1	—	—	—	—	—	—	—	—	—	—	—	—	—	—
00011b	GPT ²	—	—	GTCP O13	GTIOC 11A	GTIOC 11B	GTCP O13	GTIOC 13B	GTIOC 13A	GTIOC 12B	GTIOC 12A	GTCP O12	GTIOC 3B	GTIOC 3A	GTCP O11	GTIOC 5B	GTIOC 5A
00100b	SCI ¹	—	—	—	—	—	—	CTS6_ A	SCK6_ A	TXD6_ A/ SDA6_ A/ MOSI6_ A	RXD6_ A/ SCL6_ A/ MISO6_ A	—	—	—	—	CTS_R TS6_B/ SS6_B	CTS6_ B
00101b	SCI ¹	CTS3_ C	CTS_R TS3_C/ SS3_C	—	—	—	—	RXD3_ B/ SCL3_ B/ MISO3_ B	—	—	—	—	—	—	—	—	—
01001b	RTC/ CLKOUT/ ACMPHS/ ETHPHYC LK	—	—	ETHPH YCLK	—	—	—	—	—	—	—	—	—	—	—	CLKOU T	—
01010b	CAC/ ADC16H	—	—	—	—	—	AD1FL AG1	AD0FL AG1	ADSYN C	ADST1	ADST0	—	—	—	—	—	—
01011b	BUS	—	—	ALE	—	—	—	—	—	—	—	—	—	—	—	—	—
01101b	SCI	—	—	—	—	—	—	—	DE6	—	—	—	—	—	—	DE6	—
01110b	SCI	—	DE3	—	—	—	—	—	—	—	—	—	—	—	—	—	—
10000b	CANFD	—	—	—	—	—	—	—	—	CRX1	CTX1	—	—	—	—	—	—
10011b	USBFS	—	—	—	—	—	—	USB_ID	USB_E XICEN	USB_O VRCUR B	USB_O VRCUR A	—	—	—	—	—	—
10110b	ESWM (GMII/MII)	—	—	—	—	ET1_R XD4	ET1_R X_CLK	ET1_R XD0	ET1_R XD1	ET1_R XD2	ET1_R XD3	ET1_T XD4	ET1_T XD5	ET1_T XD6	ET1_T XD7	—	—
10111b	ESWM (RMII)	—	—	—	—	—	RMII1_ REF50 CK	RMII1_ RXD0	RMII1_ RXD1	—	—	—	—	—	—	—	—
11000b	ESWM (RGMII)	—	—	—	—	—	RGMII1_ RXC	RGMII1_ RXD0	RGMII1_ RXD1	RGMII1_ RXD2	RGMII1_ RXD3	—	—	—	—	—	—
11010b	ESC	—	—	—	—	—	CAT1_ RX_CL K	CAT1_ ERXD0	CAT1_ ERXD1	CAT1_ ERXD2	CAT1_ ERXD3	—	—	—	—	—	—
11110b	ULPT	—	—	—	—	—	—	ULPTO 1	ULPTE E1	ULPTE V1	ULPTO A1	—	—	—	—	—	—
ASEL bit		—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
ISEL bit		IRQ30	IRQ31	IRQ0	IRQ1	IRQ2	IRQ8	IRQ9	IRQ10	IRQ11	IRQ21- DS	IRQ7	IRQ6	IRQ5	IRQ3	IRQ9	IRQ8
PCR bit	Pull-up	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
NCODR bit	N-ch Open-drain	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
DSCR[1:0] bit	Drive capability control	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH
289, 303 pins product		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
224 pins product		—	—	✓	—	—	✓	✓	✓	✓	✓	—	—	—	—	—	—
176 pins product		—	—	—	—	—	✓	✓	✓	✓	✓	—	—	—	—	—	—

✓: Available

—: Setting prohibited

Note 1. Recommend using pins that have a letter appended to their names, for instance “_A”, “_B”, or “_C” to indicate group membership. For the interface, the AC portion of the electrical characteristics is measured for each group.

Note 2. There are 2 types output buffer which are middle drive and high drive. Recommend using same drive buffer for output skew spec (tGTISK).

Table 20.17 Register settings for input/output pin function (PORTA)

PSEL [4:0]Settings	Function	Pin															
		PA00	PA01	PA02	PA03	PA04	PA05	PA06	PA07	PA08	PA09	PA10	PA11	PA12	PA13	PA14	PA15
00000b (value after reset)	Hi-z/ JTAG/SWD	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z
00010b	GPT ²	GTOVLO	GTOVUP	GTIWP	GTIV	GTIU	GTETRGD	GTETRG	GTETRGB	GTETRGD	—	—	GTIV	GTIW	GTOVUP	GTOVLO	—
00011b	GPT ²	GTCPP07	GTCPP08	GTCPP09	GTCPP010	GTIOC4B	GTIOC4A	GTIOC7B	GTIOC7A	GTCPP011	GTCPP012	GTCPP013	GTIOC6A	GTIOC6B	GTIOC10A	GTIOC10B	GTIOC7A
00100b	SCI ¹	—	—	RXD2_C/ SCL2_C/ MISO2_C	TXD2_C/ SDA2_C/ MOSI2_C	SCK2_C	CTS_RTS2_C/ SS2_C	CTS2_C	—	—	—	—	—	—	—	—	—
00101b	SCI ¹	CTS_RTS5_B/ SS5_B	CTS5_B	—	—	—	—	—	CTS7_A	RXD5_B/ SCL5_B/ MISO5_B	TXD5_B/ SDA5_B/ MOSI5_B	SCK5_B	SCK9_C	RXD9_C/ SCL9_C/ MISO9_C	CTS_RTS9_C/ SS9_C	TXD9_C/ SDA9_C/ MOSI9_C	CTS9_C
01001b	RTC/ CLKOUT/ ACMPHS/ ETHPHYCLK	—	—	—	—	—	—	—	VCOUT	—	—	—	—	—	—	—	—
01010b	CAC/ ADC16H	AD1FLAG1	AD0FLAG1	ADSYN C	ADST1	ADST0	—	—	—	—	—	—	—	—	—	—	—
01011b	BUS	A5	A4	A3	A2	A1/ DQM3	A0/BC0/ DQM1	CS1/CKE	RD	CS0/WE	CS3/CAS	CS2/RAS	D8/DQ8	D9/DQ9	D10/ DQ10	D11/ DQ11	EBCLK/ SDCLK
01101b	SCI	—	—	—	—	DE2	DE2	—	—	—	—	—	—	—	—	—	—
01110b	SCI	DE5	—	—	—	—	—	—	—	—	—	DE5	DE9	—	DE9	—	—
10101b	SDHI ¹	SD0DAT7_A	SD0DAT6_A	SD0DAT5_A	SD0DAT4_A	SD0DAT3_A	SD0DAT2_A	SD0DAT1_A	SD0DAT0_A	—	—	—	—	—	—	—	—
ASEL bit		—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
ISEL bit		IRQ22	IRQ21	IRQ31	IRQ20	IRQ19	IRQ18	IRQ17	IRQ16	IRQ6	IRQ5	IRQ4	IRQ10	IRQ11	IRQ12	IRQ13	IRQ14
PCR bit	Pull-up	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
NCODR bit	N-ch Open-drain	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
DSCR[1:0] bit	Drive capability control	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH	L/M/H/ HH
289, 303 pins product		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
224 pins product		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
176 pins product		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

✓: Available
 —: Setting prohibited

Note 1. Recommend using pins that have a letter appended to their names, for instance “_A”, “_B”, or “_C” to indicate group membership. For the interface, the AC portion of the electrical characteristics is measured for each group.

Note 2. There are 2 types output buffer which are middle drive and high drive. Recommend using same drive buffer for output skew spec (t_{GTISK}).

Table 20.18 Register settings for input/output pin function (PORTB) (1 of 2)

PSEL [4:0]Settings	Function	Pin															
		PB00	PB01	PB02	PB03	PB04	PB05	PB06	PB07	PB08	PB09	PB10	PB11	PB12	PB13	PB14	
00000b (value after reset)	Hi-z/ JTAG/SWD	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	
00011b	GPT ²	GTCPP O4	GTCPP O2	GTCPP O0	GTCPP O1	GTCPP O3	GTCPP O4	GTIOC9 A	GTIOC9 B	—	—	—	—	—	—	—	
00101b	SCI ¹	SCK1_B	CTS_RT S1_B/ SS1_B	RXD5_C / SCL5_C / MISO5_C	TXD5_C / SDA5_C / MOSI5_C	SCK5_C	CTS5_C	CTS_RT S5_C/ SS5_C	—	—	—	—	—	—	—	—	
01000b	DSMIF	DSM1C LK2	—	—	—	—	—	—	—	—	—	—	—	—	—	—	
01010b	CAC/ ADC16H	ADST0	AD1FLA G1	ADST1	ADSYN C	AD0FLA G1	—	—	—	—	—	—	—	—	—	—	

Table 20.18 Register settings for input/output pin function (PORTB) (2 of 2)

PSEL [4:0]Settings	Function	Pin														
		PB00	PB01	PB02	PB03	PB04	PB05	PB06	PB07	PB08	PB09	PB10	PB11	PB12	PB13	PB14
01011b	BUS	—	ALE	—	—	—	—	—	—	—	—	—	—	—	—	—
01110b	SCI	DE1	DE1	—	—	DE5	—	DE5	—	—	—	—	—	—	—	—
10110b	ESWM (GMII/MII)	ET0_TX D0	ET0_TX _CLK	ET0_TX D1	ET0_TX D2	ET0_TX D3	ET0_TX D7	ET0_TX D6	ET0_TX D5	—	—	—	—	—	—	—
10111b	ESWM (RMII)	RMII0_T XD0	—	RMII0_T XD1	—	—	—	—	—	—	—	—	—	—	—	—
11000b	ESWM (RGMII)	RGMII0_ TXD0	—	RGMII0_ TXD1	RGMII0_ TXD2	RGMII0_ TXD3	—	—	—	—	—	—	—	—	—	—
11010b	ESC	CAT0_E TXD0	CAT0_T X_CLK	CAT0_E TXD1	CAT0_E TXD2	CAT0_E TXD3	—	—	—	CATLED RUN	CATIRQ	CATLED STER	CATLED ERR	CATLIN KACT0	CATLIN KACT1	CATSY NC0
ASEL bit		—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
ISEL bit		IRQ10	IRQ12	IRQ11	IRQ13	IRQ9	IRQ15	IRQ0	IRQ1	IRQ12	IRQ13	IRQ4	IRQ5	IRQ6	IRQ7	IRQ10
PCR bit	Pull-up	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
NCODR bit	N-ch Open- drain	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
DSCR[1:0] bit	Drive capability control	L/M/H/H H	L/M/H/H H	L/M/H/H H	L/M/H/H H	L/M/H/H H	L/M/H/H H	L/M/H/H H	L/M/H/H H	L/M/H/H H	L/M/H/H H	L/M/H/H H	L/M/H/H H	L/M/H/H H	L/M/H/H H	L/M/H/H H
289, 303 pins product		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
224 pins product		✓	✓	✓	✓	✓	—	—	—	✓	✓	✓	✓	✓	✓	✓
176 pins product		✓	✓	✓	✓	✓	—	—	—	✓	✓	✓	✓	✓	✓	✓

✓: Available

—: Setting prohibited

Note 1. Recommend using pins that have a letter appended to their names, for instance “_A”, “_B”, or “_C” to indicate group membership. For the interface, the AC portion of the electrical characteristics is measured for each group.

Note 2. There are 2 types output buffer which are middle drive and high drive. Recommend using same drive buffer for output skew spec (t_{GTISK}).

Table 20.19 Register settings for input/output pin function (PORTC) (1 of 2)

PSEL [4:0]Settings	Function	Pin															
		PC00	PC01	PC02	PC03	PC04	PC05	PC06	PC07	PC08	PC09	PC10	PC11	PC12	PC13	PC14	PC15
00000b (value after reset)	Hi-z/ JTAG/SW D	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z
00010b	GPT ²	—	—	—	—	—	—	—	—	—	—	—	—	—	—	GTADS M1	GTADS M0
00011b	GPT ²	GTCPP O7	GTCPP O6	GTCPP O5	GTCPP O4	GTCPP O3	GTCPP O2	GTCPP O1	GTCPP O0	GTCPP O8	—	GTCPP O13	GTCPP O12	GTCPP O11	GTCPP O10	GTCPP O9	—
00100b	SCI ¹	—	—	—	—	—	—	—	—	—	—	—	CTS_R TS6_C/ SS6_C	SCK6_ C	RXD6_ C/ SCL6_ C/ MISO6_ C	TXD6_ C/ SDA6_ C/ MOSI6_ C	CTS6_ C
00101b	SCI ¹	CTS_R TS7_C/ SS7_C	CTS7_ C	SCK7_ C	TXD7_ C/ SDA7_ C/ MOSI7_ C	RXD7_ C/ SCL7_ C/ MISO7_ C	—	—	—	—	—	—	—	—	—	—	—
01000b	DSMIF	—	—	DSM0C LK2	DSM0C LK1	DSM0C LK0	DSM0D AT2	DSM0D AT1	DSM0D AT0	DSM1D AT0	—	DSM1C LK2	DSM1C LK1	DSM1C LK0	DSM1D AT2	DSM1D AT1	—
01011b	BUS	D30/ DQ30	D29/ DQ29	D28/ DQ28	D27/ DQ27	D26/ DQ26	D25/ DQ25	D24/ DQ24	D23/ DQ23	D22/ DQ22	D21/ DQ21	D20/ DQ20	D19/ DQ19	D18/ DQ18	D17/ DQ17	D16/ DQ16	A16
01101b	SCI	—	—	—	—	—	—	—	—	—	—	—	DE6	DE6	—	—	—
01110b	SCI	DE7	—	DE7	—	—	—	—	—	—	—	—	—	—	—	—	—
10000b	CANFD	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CRX1
10110b	ESWM (GMII/MII)	—	—	—	—	—	—	—	—	—	—	—	ET0_M DC	ET0_M DIO	ET0_IN T	ET0_W OL	—
11010b	ESC	—	—	—	—	—	—	—	—	—	—	—	CAT0_ MDC	CAT0_ MDIO	—	—	—
11100b	OSPI	OM_1_ SIO5	OM_1_ SIO0	OM_1_ SIO3	OM_1_ SIO4	OM_1_ SIO2	OM_1_ CS1	OM_1_ ECSIN T1	OM_1_ RESET	OM_1_ CS0	OM_1_ RSTO1	OM_1_ WP1	—	—	—	—	—

Table 20.19 Register settings for input/output pin function (PORTC) (2 of 2)

PSEL [4:0] Settings	Function	Pin															
		PC00	PC01	PC02	PC03	PC04	PC05	PC06	PC07	PC08	PC09	PC10	PC11	PC12	PC13	PC14	PC15
ASEL bit		—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
ISEL bit		IRQ28	IRQ27	IRQ26	IRQ25	IRQ24	IRQ23	IRQ22	IRQ21	IRQ20	IRQ19	IRQ18	IRQ17	IRQ16	IRQ15	IRQ14	IRQ13
PCR bit	Pull-up	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
NCODR bit	N-ch Open-drain	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
DSCR[1:0] bit	Drive capability control	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH
289, 303 pins product ³		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
224 pins product		—	—	—	—	—	—	—	—	—	—	—	✓	✓	✓	✓	✓
176 pins product		—	—	—	—	—	—	—	—	—	—	—	✓	✓	✓	✓	✓

✓: Available

—: Setting prohibited

Note 1. Recommend using pins that have a letter appended to their names, for instance “_A”, “_B”, or “_C” to indicate group membership. For the interface, the AC portion of the electrical characteristics is measured for each group.

Note 2. There are 2 types output buffer which are middle drive and high drive. Recommend using same drive buffer for output skew spec (t_{GTISK}).

Note 3. For BGA303 package, PC00 to PC07 are connected to the SiP Flash memory inside the package. To access SiP Flash memory, set OSPI in PSEL settings.

Table 20.20 Register settings for input/output pin function (PORTD)

PSEL[4:0] Settings	Function	Pin							
		PD00	PD01	PD02	PD03	PD04	PD05	PD06	PD07
00000b (value after reset)	Hi-z/JTAG/SWD	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z	Hi-z
00010b	GPT ²	GTOWLO	—	—	—	—	—	—	—
00011b	GPT ²	GTCPPO5	GTCPPO2	GTCPPO1	GTIOC3B	GTIOC3A	GTIOC2B	GTIOC2A	GTCPPO0
00100b	SCI ¹	—	SCK8_C	TXD8_C/ SDA8_C/ MOSI8_C	RXD8_C/ SCL8_C/ MISO8_C	CTS_RTS8_C/ SS8_C	CTS8_C	—	—
00101b	SCI ¹	SCK5_A	—	—	—	—	—	—	—
01011b	BUS	A15	—	—	—	—	—	—	—
01101b	SCI	—	DE8	—	—	DE8	—	—	—
01110b	SCI	DE5	—	—	—	—	—	—	—
10000b	CANFD	CTX1	—	—	—	—	—	—	—
10101b	SDHI ¹	—	SD0DAT2_C	SD0DAT1_C	SD0DAT0_C	SD0CMD_C	SD0CLK_C	SD0WP	SD0CD
10110b	ESWM (GMII/MII)	—	ET1_RXD6	ET1_RXD7	ET0_RXD4	ET0_RXD5	ET0_RXD6	ET0_RXD7	ET0_TXD4
ASEL bit		—	—	—	—	—	—	—	—
ISEL bit		IRQ23	IRQ22	IRQ21	IRQ21	IRQ20	IRQ19	IRQ18	IRQ17
PCR bit	Pull-up	✓	✓	✓	✓	✓	✓	✓	✓
NCODR bit	N-ch Open-drain	✓	✓	✓	✓	✓	✓	✓	✓
DSCR[1:0] bit	Drive capability control	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH	L/M/H/HH
289, 303 pins product		✓	✓	✓	✓	✓	✓	✓	✓
224 pins product		✓	—	—	—	—	—	—	—
176 pins product		✓	—	—	—	—	—	—	—

✓: Available

—: Setting prohibited

Note 1. Recommend using pins that have a letter appended to their names, for instance “_A”, “_B”, or “_C” to indicate group membership. For the interface, the AC portion of the electrical characteristics is measured for each group.

Note 2. There are 2 types output buffer which are middle drive and high drive. Recommend using same drive buffer for output skew spec (t_{GTISK}).